



Research paper

A linear-complexity deep learning framework for efficient battery State of Health estimation in resource-constrained systems[☆]Hua Jiang Wang^{a,} , Hao Li^a, Zhengxing He^a, Xuan Chen^a, Hanglin Liu^a, Chao Chen^{a, b,} , *^a School of Computer Science and Engineering, Sichuan University of Science and Engineering, Yibin 644005, China^b Intelligent Perception and Control Key Laboratory of Sichuan Province, Sichuan University of Science and Engineering, Yibin 644005, China

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ABSTRACT

To address the computational bottleneck of deep learning, a key artificial intelligence (AI) technology, in resource-constrained Battery Management Systems (BMS), this paper presents a lightweight, linear-complexity State of Health (SOH) estimation framework. First, a systematic feature selection process identifies robust health indicators from partial voltage-time integrals, effectively reducing online extraction burdens. Subsequently, we propose SL-AgentNet (integrating Small-kernel and Large-kernel convolutions), an AI architecture designed to resolve the quadratic overhead of traditional Transformers. By integrating the Fast Local-Global Fusion Attention (FLFA) mechanism with diversity-enhancing modules, specifically the Fast Diversity Feature Fusion Attention (FDFA) and Parallel Informed Activation Function (PIAF), this AI model achieves linear complexity while capturing long-range degradation dependencies. Experiments on the Oxford, National Aeronautics and Space Administration (NASA), and Center for Advanced Life Cycle Engineering (CALCE) datasets demonstrate superior robustness, reducing the Mean Absolute Error (MAE) by 74.78% on the Oxford dataset against Convolutional Neural Network (CNN)-Transformer baselines. With a model size of only 40.92 Kilobytes (KB), the AI framework proves its feasibility through hardware-in-the-loop validation on a Texas Instruments (TI) Cortex-M4 microcontroller, where a Look-Up Table (LUT) implementation maintains accuracy loss within 0.3%, demonstrating its readiness for embedded deployment.

1. Introduction

Global energy sustainability relies heavily on critical advancements across the broad energy storage and management spectrum. Recent years have witnessed diversified progress in this domain. In the realm of novel materials, significant strides have been made in stabilizing alternative chemistries, such as the interface synergistic stabilization for zinc anodes (Hu et al., 2024) and movable SiO₂ reinforced coatings (Wu et al., 2025). Simultaneously, at the grid system level, new source-load coordinated optimization frameworks have been proposed to enhance distributed energy efficiency (Wang et al., 2025). Concurrently with these emerging technologies, Lithium-ion batteries (LIBs) currently remain the primary power source for Electric Vehicles (EVs). However, as LIB technology evolves towards higher energy densities (e.g., high-nickel cathodes, cell-to-pack (CTP) architectures),

they exhibit highly non-linear and heterogeneous degradation behaviors. These complexities challenge the fidelity of traditional physical models, making accurate State of Health (SOH) monitoring within resource-constrained Battery Management Systems (BMS) increasingly challenging (Rouholamini et al., 2022; Krause et al., 2021).

Accurate estimation of SOH is crucial for mitigating range anxiety and reducing the risk of potential failures. While direct measurement of capacity is typically considered accurate, it requires time-consuming full charge-discharge cycles, a requirement often impractical for on-line applications (Wei et al., 2018). Consequently, data-driven estimation methods that infer SOH from indirect sensor data have gained prominence. Researchers have extensively investigated various Health Indicators (HIs), such as incremental capacity (IC) curves, differential voltage analysis, and voltage integral areas, to characterize degradation (Wu et al., 2023; Feng and Zhang, 2024). However, these

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features may exhibit sensitivity to measurement noise or necessitate complex preprocessing. Early research predominantly focused on empirical models or standard machine learning. For instance, decision tree approaches were explored for fault detection during power swings (Mohanty et al., 2023), and enhanced protective relaying schemes were designed for compensated transmission lines (Mohanty et al., 2024). As highlighted in these studies across the broader energy sector, traditional methods often face a common constraint: the need to balance algorithmic sensitivity with computational efficiency under dynamic operating conditions.

The proliferation of deep learning has facilitated the widespread adoption of Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) to capture complex aging dynamics. Nevertheless, CNNs are limited by their local receptive fields, while RNNs suffer from sequential processing constraints that hinder parallelization. While the Transformer architecture (Vaswani et al., 2017) excels at capturing long-term dependencies, it can introduce a bottleneck: standard self-attention entails quadratic computational complexity $O(N^2)$ with respect to sequence length.

This substantial computational overhead is often challenging for automotive-grade microcontrollers (MCUs) (Xiong et al., 2019). Furthermore, deep Transformer models may be prone to the rank collapse issue (Chen, 2023), a phenomenon where output representations become identical across tokens. This convergence may lead to the loss of fine-grained local degradation details critical for early fault detection.

To address these challenges, this paper proposes SL-AgentNet, a lightweight and efficient SOH estimation framework. In contrast to many existing methods that pursue structural complexity, the proposed approach aims to balance engineering feasibility with accuracy. The main contributions are:

1. **Structured Feature Selection Framework:** We propose a data-driven screening methodology based on the Constant Current Charging/Discharging Area (CCCA/CCDA). In contrast to static feature selection, this framework treats the observation window as a learnable hyperparameter, adaptively identifying a dataset-specific optimized interval. This facilitates the extraction of a robust Health Indicator (HI) that reduces the online computation burden while enhancing correlation with capacity fade.
2. **Integrated Lightweight Architecture:** We introduce the SL-AgentNet, featuring a tri-branch design. The core Fast Local-Global Fusion Attention (FLFA) reduces complexity from quadratic $O(N^2)$ to linear $O(N)$ by employing a single-agent mechanism ($n = 1$) with convolutional compensation. To mitigate potential feature collapse, we integrate the Fast Diversity Feature Fusion Attention (FDFA) for orthogonal significance filtering and the Parallel Informed Activation Function (PIAF) for heterogeneous non-linear mapping. This architecture aims to achieve enhanced representational capability with a compact parameter set.
3. **Validation of Feasibility and Reliability:** Comprehensive experiments on the Oxford, NASA, and CALCE datasets demonstrate that SL-AgentNet achieves competitive accuracy, exemplified by an RMSE of 0.0052 on NASA, even when trained on 30% of early-life data. Furthermore, we provide a rigorous analysis of computational efficiency and uncertainty quantification. Validated on a TI Cortex-M4 MCU, the proposed LUT-based implementation achieves negligible precision loss ($< 0.3\%$), proving its feasibility for resource-constrained embedded instrumentation.

2. Related work

Methodologies for battery SOH estimation can be classified into model-based and data-driven approaches Xie et al. (2024). This section reviews representative techniques in each category and analyzes their specific limitations within the context of vehicular embedded applications.

2.1. Model-based approaches

Model-based methods are based on mathematical representations of the battery's internal electrochemical or electrical dynamics.

- **Electrochemical Models (EMs):** Models such as the Single Particle Model (SPM) and Pseudo-Two-Dimensional (P2D) model describe internal degradation mechanisms, including solid electrolyte interphase (SEI) growth and lithium plating, utilizing systems of Partial Differential Equations (PDEs) (Birkel et al., 2017). While offering high fidelity and physical interpretability, their high computational complexity and the difficulty of identifying numerous microscopic parameters online may render them less suitable for low-cost, resource-constrained BMS MCUs (Chaturvedi et al., 2010; Reckemmer et al., 2019).
- **Equivalent Circuit Models (ECMs):** ECMs simulate battery dynamics using electrical components, such as resistors and capacitors, and are frequently combined with adaptive filters like Kalman Filters (KF) or Particle Filters (PF) (Zhang et al., 2018; Peng et al., 2025). Although computationally lighter than EMs, their accuracy depends on the fidelity of the circuit structure. Furthermore, ECM parameters exhibit high sensitivity to environmental conditions, including temperature and C-rate (charge/discharge rate), and maintaining robustness across the full aging lifespan remains a significant challenge (Luo et al., 2022).

2.2. Data-driven approaches

Data-driven methods treat the battery as a black box, mapping extracted features directly to SOH. These methods have gained prominence due to their flexibility Richardson et al. (2017) and decoupling from complex physical parameters. Recent work has also emphasized security considerations in battery state estimation Tian et al. (2023).

2.2.1. Feature engineering: Health indicators (HIs)

The efficacy of data-driven models is closely linked to the quality of input features.

- **Differential Analysis:** Incremental Capacity Analysis (ICA) and Differential Voltage Analysis (DVA) are widely employed to identify degradation modes (Z. Kong et al., 2021). Alternative approaches exploit differential temperature Tian et al. (2020) or entropy-based metrics Widodo et al. (2011) to characterize degradation. However, mathematically, calculating derivatives (dQ/dV or dV/dQ) from discrete sampling points can act as a high-pass filter, which may amplify sensor noise. Consequently, these methods require heavy smoothing preprocessing, introducing latency and computational overhead.
- **Raw Data and Geometric Features:** While utilizing raw charging curves directly avoids feature engineering, it may often lead to input redundancy. Geometric features, such as the time intervals of equal discharge voltage difference (TIEDVD) (Liu et al., 2013), voltage integral areas (Zhang et al., 2024a), and novel health indicators Zhou et al. (2016) offer enhanced stability. Multi-feature fusion strategies have also been widely explored Lin et al. (2022), Son et al. (2022), with recent works investigating automated feature selection Li and Chen (2025). However, prior works typically employ empirically fixed voltage windows, often lacking a systematic mechanism to adapt to the specific aging fingerprints of different battery chemistries, such as Nickel Cobalt Manganese (NCM) versus Lithium Iron Phosphate (LFP).

2.2.2. Deep learning architectures

Deep learning has transformed SOH estimation by enabling end-to-end feature learning.

- **RNNs and CNNs:** Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks, represent the standard for time-series forecasting (Ma et al., 2019; Zheng et al., 2023). Despite their success, the sequential nature of RNNs limits parallelization during training and inference, limiting throughput. Convolutional Neural Networks (CNNs) (Lee et al., 2023) facilitate parallel processing but are constrained by local receptive fields, making it difficult to capture global degradation trends over long operational cycles. Hybrid CNN-based frameworks have been combined with random forests Yang et al. (2022), while ensemble learning paradigms further enhance estimation robustness Zhang et al. (2024b). Additionally, LSTM-based models leveraging incremental capacity features Zhang et al. (2023) remain a strong baseline for battery health prognostics.
- **Transformers and Attention Mechanisms:** The introduction of the Transformer architecture (Hannan et al., 2021) addressed the limitations of RNNs and CNNs by utilizing self-attention to model long-range dependencies. Hybrid models, such as the CNN-Transformer (Bai and Wang, 2023; Gu et al., 2023) and BiGRU-Transformer Jia et al. (2023), have further improved accuracy. Vision Transformer Chen et al. (2023) and temporal fusion Transformer Gomez et al. (2024) architectures have also been explored for battery health estimation. More recently, efficient Transformer variants such as BMSformer Li et al. (2024) have been specifically designed for online SOH estimation. However, the standard Dot-Product Attention mechanism involves a time and memory complexity of $O(N^2)$ with respect to sequence length N . This quadratic cost presents a bottleneck for deployment on edge devices (Xiong et al., 2019). Moreover, as noted in recent theoretical studies (Chen, 2023), deep attention layers may be prone to rank collapse, a phenomenon where the output feature matrix loses expressiveness, limiting the model's ability to distinguish fine-grained aging patterns.

To address these limitations, this study introduces SL-AgentNet. By replacing quadratic attention with a linear-complexity $O(N)$ Agent Attention mechanism and integrating systematic feature selection, our framework aims to address the trilemma of noise sensitivity, computational burden, and feature expressiveness.

3. Methodology framework and data foundation

3.1. Overall estimation framework

SOH is a key indicator of battery degradation, quantifying the ability of a battery to store and deliver energy relative to its fresh state. While universal standardization remains absent, capacity retention persists as the most widely adopted operational proxy (Lu et al., 2023). In this study, SOH is defined as the ratio of the current maximum available capacity (C_{current}) to the rated nominal capacity (C_{rated}):

$$\text{SOH} = \frac{C_{\text{current}}}{C_{\text{rated}}} \times 100\% \quad (1)$$

Although SOH can also be characterized by the increase in internal resistance, this study primarily focuses on the capacity fade dimension as the primary metric for energy storage capability.

Given that direct capacity measurement is often impractical for online BMS due to the necessity of full charge–discharge cycles, a data-driven framework, depicted in Fig. 1, is proposed. This systematic methodology comprises four strategic phases:

1. **Data Acquisition:** This phase entails extracting raw voltage, current, and time series data from charging/discharging cycles across diverse battery chemistries.

2. **Systematic Feature Engineering:** A multi-scale sliding window strategy is applied to identify robust Health Indicators (HIs) from partial voltage segments, as detailed in Section 3.3.
3. **Model Development:** The proposed SL-AgentNet is trained using a historical reference hyperparameter calibration strategy to promote efficiency and accuracy.
4. **Performance Evaluation:** The final phase validates the model's estimation precision, computational efficiency, and robustness against baselines on unseen test cells.

3.2. Experimental datasets

To evaluate the generalization capability of the proposed method across varying chemistries and operational profiles, three authoritative public datasets were utilized. The specifications are summarized as follows:

- **Oxford Dataset (Pouch, Lithium Cobalt Oxide (LCO)-based):** Provided by the Oxford Battery Intelligence Lab (Birkel et al., 2017), this dataset includes 8 pouch cells (0.74 Ah) tested at 40 °C. It is worth noting that it employs the Urban Artemis drive cycle for the discharging phase. Although our Health Indicators are primarily derived from the stable constant-current charging phase, this dynamic discharging profile induces realistic degradation patterns distinct from standard constant-current cycling, thereby making it a rigorous benchmark for evaluating model robustness under dynamic load history.
- **NASA Dataset (18650, Nickel Cobalt Aluminum (NCA)):** Sourced from the NASA Prognostics Center of Excellence (Xu et al.), this dataset includes 4 cylindrical cells (B0005–B0018) with a rated capacity of 2.0 Ah, cycled at 24 °C. A distinct characteristic of this dataset is the significant capacity regeneration phenomena, manifested as local fluctuations, and varying discharge currents. These features provide a rigorous test for the model's stability and temporal feature extraction capabilities.
- **CALCE Dataset (Prismatic, LCO):** Provided by the Center for Advanced Life Cycle Engineering (CALCE) at the University of Maryland Cadini et al. (2019), this dataset features 4 prismatic cells (CS2_35 to CS2_38) with a rated capacity of 1.1 Ah. These batteries exhibit a rapid non-linear degradation tail, or “diving” behavior, in late life stages, testing the model's ability to track accelerated aging trends.

The diversity in cell formats (Pouch/Cylindrical/Prismatic), rated capacities (0.74–2.0 Ah), and cathode chemistries (LCO-based/NCA) facilitates a comprehensive validation of the proposed framework.

3.3. Systematic extraction and screening of health indicators

3.3.1. Theoretical advantage of CCCA/CCDA over energy metrics

Conventional energy-based methods (Bai et al., 2023) characterize battery aging by calculating the total energy throughput during a cycle:

$$E = \int_{t_{\text{start}}}^{t_{\text{end}}} V(t) \cdot I(t) dt \quad (2)$$

However, current sensors in practical BMS deployments—typically Hall effect sensors or shunt resistors—can be susceptible to measurement noise, thermal drift, and instantaneous load fluctuations. Since energy calculation involves the multiplicative coupling of voltage and current, any jitter in the current signal $I(t)$ is potentially amplified and integrated over time, resulting in cumulative errors in the final SOH estimate.

To circumvent reliance on noisy current data, the utilization of the Constant Current Charging Area (CCCA) and Constant Current Discharging Area (CCDA) is proposed. In the Constant Current (CC) phase, the current is theoretically invariant ($I(t) \approx I_{\text{set}}$). Consequently,

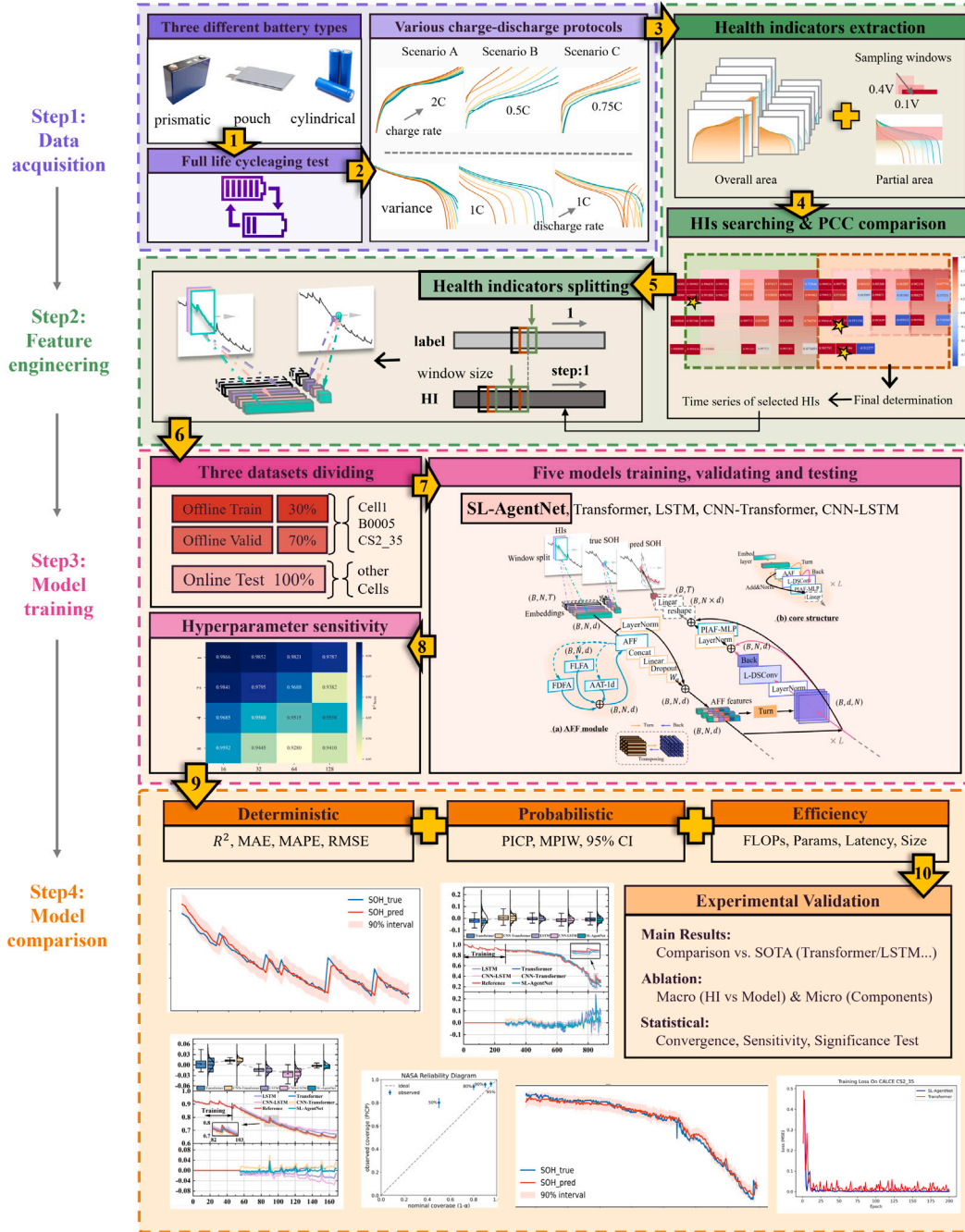


Fig. 1. Schematic diagram of the proposed SOH estimation methodology, illustrating the workflow from raw data acquisition to final performance evaluation.

capacity degradation—manifested as the shortening of the CC phase duration and the shifting of voltage plateaus—can be effectively captured by the geometric integral of the voltage curve (as visualized in Fig. 2):

$$CCCA_i = \int_{t_{start}}^{t_{end}} V_i(t) dt \quad (3)$$

Operationally, CCCA serves as a pseudo-energy metric with enhanced noise resilience. By decoupling the health indicator from real-time current measurements, this metric retains the high sensitivity of energy integration to degradation while offering robustness against sensor noise, thereby offering enhanced stability for online instrumentation.

Furthermore, unlike energy metrics which conflate charging and discharging behaviors, our method allows for Kinetic Decoupling. By treating CCCA and CCDA as independent candidates, we can isolate the specific kinetic regime that maximizes the signal-to-noise ratio (SNR)

for a given chemistry, thereby enhancing the “Information Density” of the input feature.

3.3.2. Multi-scale sliding window screening strategy

Although voltage integration is established in the literature, prior studies typically rely on static, empirically defined windows. Battery degradation mechanisms, however, are heterogeneous; consequently, the voltage region containing the most salient aging information varies significantly across chemistries.

To address this limitation, the Systematic Screening Framework is introduced (Fig. 3). To adhere to prognostic standards and minimize the risk of data leakage, this process is designed as an offline calibration step, performed exclusively on available historical reference data (i.e., the training partition) prior to model deployment. The procedure involves three rigorous steps:

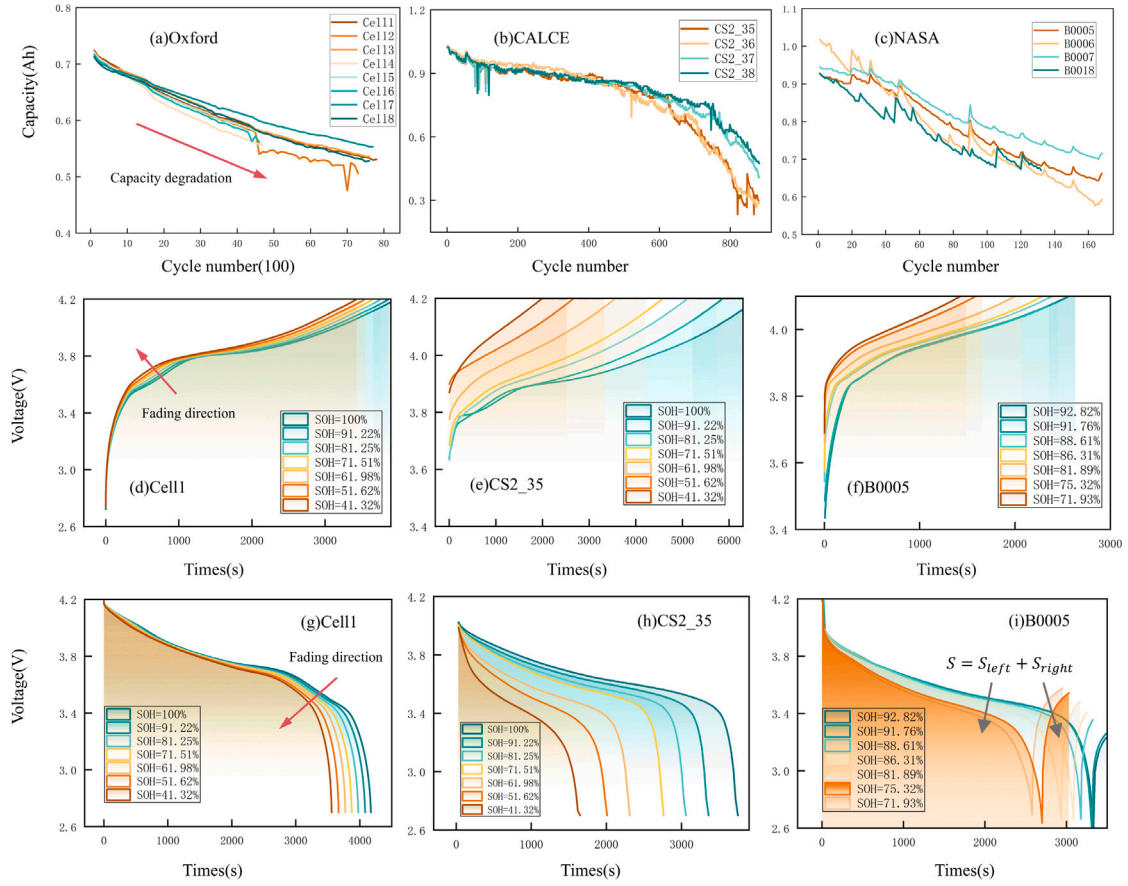


Fig. 2. Visualization of the experimental datasets. (a)–(c) Capacity degradation curves (SOH ground truth) for Oxford, CALCE, and NASA datasets. (d)–(f) Constant current charging voltage profiles. (g)–(i) Constant current discharging voltage profiles. These curves illustrate the diverse aging patterns and measurement characteristics across different battery chemistries.

1. Data Segmentation (Multi-Scale Windows): To capture fine-grained “local fingerprints” of degradation that might be averaged out in a global integral, we systematically segment the voltage profile using sliding windows with varying granularities (e.g., 0.4 V, 0.2 V, and 0.1 V widths).

- **Coarse-grained Windows (e.g., 0.4 V):** Designed to capture broad phase transition trends and offer robustness against noise.
- **Fine-grained Windows (e.g., 0.1 V):** Designed to isolate subtle variations in specific local voltage plateaus highly sensitive to active material loss.

2. Correlation Analysis (Offline Calibration): For each candidate window defined by $[V_{low}, V_{high}]$, we calculate the integral area sequence. The Pearson Correlation Coefficient (PCC) (Roman et al., 2021; Lin et al., 2024) is computed between the feature sequence and the ground-truth SOH using only the calibration/training data:

$$PCC = \frac{\sum (x_k - \bar{x})(y_k - \bar{y})}{\sqrt{\sum (x_k - \bar{x})^2 \sum (y_k - \bar{y})^2}} \quad (4)$$

The window yielding the highest absolute PCC on this calibration set is identified as the “Optimal Interval”. The detailed PCC comparisons for feature selection across different datasets are presented in Tables 1, 2, and 3. Furthermore, Tables Table 4 and 5 benchmark the selected HI against alternative indicators from the literature, demonstrating its superior correlation with capacity degradation.

3. Feature Selection & Fixation & Deployment: Once the optimal interval is identified, it is frozen as a fixed hyperparameter for the subsequent testing and inference phases. The calibration analysis revealed distinct optimal intervals for different datasets:

- For the Oxford dataset, the full-range CCCA was identified as the optimal fixed input. This suggests a uniform capacity fade mechanism where the global integral yields a favorable signal-to-noise ratio.
- For the CALCE dataset, the calibration pointed to the partial segment (3.4–3.8 V CCDA), indicating that aging is locally concentrated in specific phase transitions for this chemistry.
- For the NASA dataset, the left section of the CCDA curve was selected. This segmentation was based on the first voltage minimum point (as depicted in Fig. 2(i)), identifying the early discharge phase as the effective indicator.

This process allows the feature extraction rules to be defined *a priori*, simulating a realistic BMS deployment where the optimal observation window is pre-configured based on battery characteristics.

3.3.3. Novelty and applicability

The core innovation of this section is the transition from static feature selection to a data-driven hyperparameter search. By treating the feature interval as a learnable hyperparameter derived during the offline calibration phase, our framework facilitates the delivery of highly informative input to the subsequent SL-AgentNet.

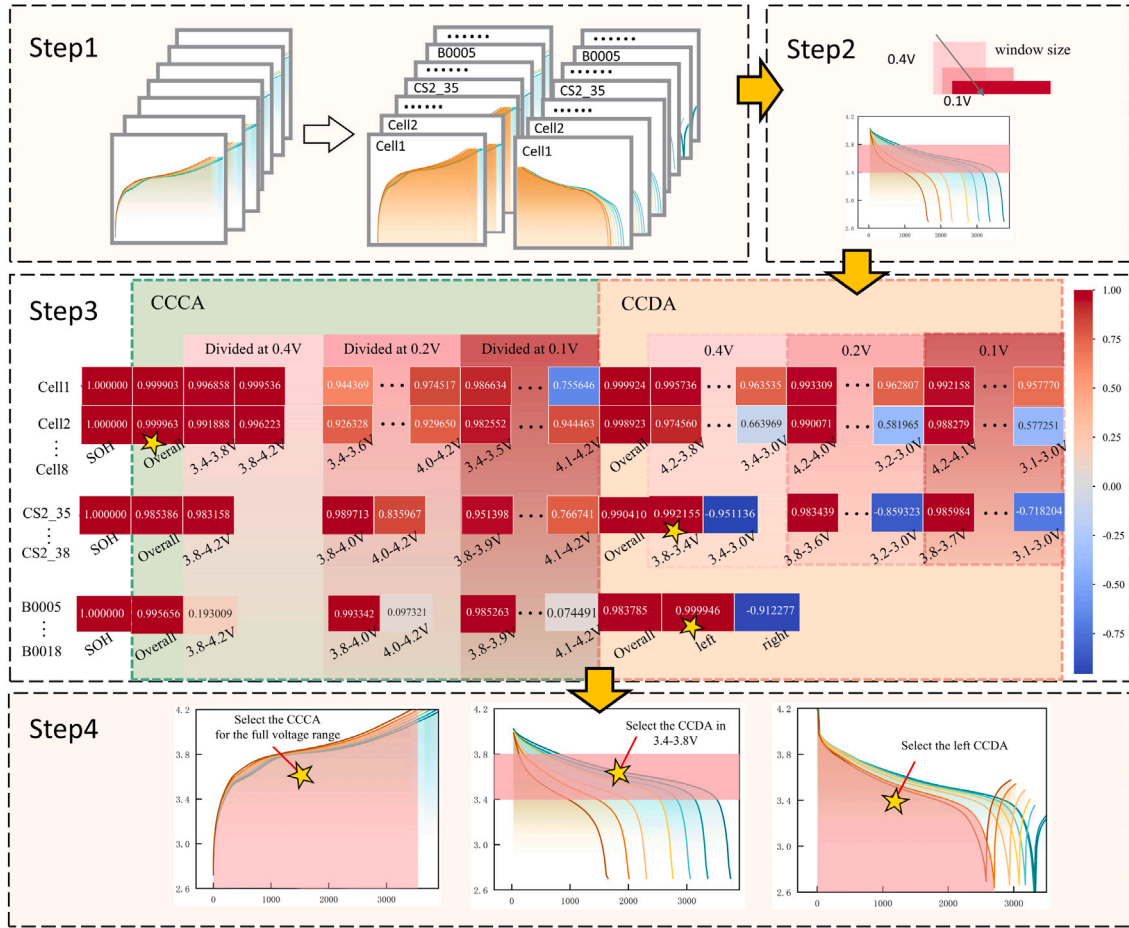


Fig. 3. Schematic of the systematic extraction procedures for selecting the optimal Health Indicator (HI) from charge/discharge segments.

Table 1

Comparison of PCC values for four candidate features on the Oxford dataset. This granular breakdown confirms the consistency of the screening method across individual cells.

Battery series	Overall CCCA	CCCA (3.8–4.2 V)	Overall CCDA	CCDA (3.8–3.4 V)
Cell1	0.999903	0.999536	0.999924	0.999499
Cell2	0.999963	0.996223	0.998923	0.991520
Cell3	0.999924	0.999500	0.999881	0.999564
Cell4	0.999973	0.998354	0.999872	0.999496
Cell5	0.999826	0.998669	0.999847	0.999463
Cell6	0.999988	0.996699	0.999062	0.998661
Cell7	0.999986	0.999126	0.999908	0.999575
Cell8	0.999969	0.998938	0.999911	0.999253
Average	0.999942	0.998381	0.999666	0.998379

Table 2

Comparison of PCC values for four candidate features on the NASA dataset.

NASA series	Overall CCCA	CCCA (3.8–4.0 V)	Overall CCDA	Left CCDA	Right CCDA
B0005	0.995656	0.995051	0.983785	0.999946	−0.981650
B0006	0.988015	0.988795	0.994266	0.999802	−0.990374
B0007	0.991336	0.986943	0.990917	0.999873	−0.785222
B0018	0.971424	0.979197	0.993277	0.999766	−0.904780
Average	0.986608	0.987497	0.990561	0.999847	−0.915507

Importantly, this approach enhances online feasibility: once the optimal interval (e.g., a 0.4 V segment) is identified offline, the BMS needs only to sample and integrate voltage within this narrow window during runtime. This effectively reduces data storage and processing requirements compared to full-curve monitoring, thereby balancing high estimation accuracy with edge-deployment constraints.

4. The proposed SL-AgentNet architecture

4.1. SL-AgentNet architecture summary

To address both the challenges of estimation accuracy and computational efficiency, a lightweight deep learning architecture named SL-AgentNet is proposed. The designation “SL” stems from its core

Table 3

Comparison of PCC values for four candidate features on the CALCE dataset.

CALCE series	Overall CCCA	CCCA (3.8–4.0 V)	Overall CCDA	CCDA (3.8–3.4 V)
CS2_35	0.985386	0.989713	0.990410	0.992155
CS2_36	0.975222	0.987095	0.990941	0.994345
CS2_37	0.979150	0.990299	0.991166	0.995010
CS2_38	0.984004	0.991994	0.991102	0.987430
Average	0.980941	0.989775	0.990905	0.992235

Table 4

Comparison of the selected HI with other HIs from literature based on PCC values (Oxford dataset).

Battery	Selected HI	CCCT (Bai et al., 2023)	CCCQ (Wu et al., 2023)	CCDQ (Bai et al., 2023)	IC Peak (Feng and Zhang, 2024)	Sample entropy (Bai et al., 2023)	Discharge slope (Feng and Zhang, 2024)
Cell1	0.99990	0.9999	0.99947	0.9998	0.9444	−0.9853	0.9999
Cell2	0.99996	0.9999	0.98677	0.9999	–	−0.9839	–
Cell3	0.99992	0.9999	0.99948	0.9989	0.9375	−0.9876	0.9998
Cell4	0.99997	0.9104	0.99981	0.9999	–	−0.9885	–
Cell5	0.99983	0.9999	0.99960	0.9999	–	−0.9133	–
Cell6	0.99999	0.9999	0.99961	0.9991	–	−0.9848	–
Cell7	0.99999	0.9999	0.99973	0.9999	0.9296	−0.9896	0.9998
Cell8	0.99997	0.9999	0.99970	0.9999	0.9378	−0.9908	0.9998

CCCT = Constant Current Charge Time.

CCCQ = Constant Current Charge Quantity.

CCDQ = Constant Current Discharge Quantity.

Table 5

Comparison of the selected HI with other HIs from literature based on PCC values (NASA and CALCE datasets).

Battery	Selected HI	CCCT (Tan and Zhao, 2020)	CCCQ (Wu et al., 2023)	Max projection distance (Jin et al., 2023)	IC Peak (Feng and Zhang, 2024)	Relax time (Huang et al., 2023)
B0005	0.99995	0.9703	0.94268	0.95	0.9910	−0.8941
B0006	0.99980	0.9366	0.94518	0.97	0.9800	−0.9373
B0007	0.99987	0.9745	0.92037	0.99	0.9251	−0.9049
B0018	0.99977	–	0.81248	–	–	–
CS2_35	0.99216	0.9799	0.90290	0.95	–	−0.9843
CS2_36	0.99435	0.9827	0.93370	0.99	–	−0.9829
CS2_37	0.99501	0.9789	0.93920	0.98	–	–
CS2_38	0.98743	0.9748	–	0.96	–	–

CCCT = Constant Current Charge Time.

CCCQ = Constant Current Charge Quantity.

CCDQ = Constant Current Discharge Quantity.

design philosophy of integrating Small-kernel and Large-kernel depthwise separable convolutions to capture multi-scale features efficiently. As illustrated in Fig. 4, the model departs from traditional parallel multi-branch designs, instead adopting a cascaded feature refinement strategy. This streamlined data flow progressively processes the input through global aggregation, local refinement, and non-linear mapping stages.

From a mathematical perspective, let the input tensor be denoted by $\mathbf{X} \in \mathbb{R}^{B \times N \times d}$, where N is the sequence length and d is the feature embedding dimension. The output of the SL-AgentNet block is computed through the following sequential operations:

Step 1: Global temporal aggregation. The input $\mathbf{X}$ is first processed by the AFF module to capture long-range dependencies. A residual connection and a learnable scaling factor $\mathbf{W}_a$ are employed to facilitate the gradient flow:

$$\mathbf{X}' = \mathbf{W}_a \cdot \text{AFF}(\text{LN}(\mathbf{X})) + \mathbf{X} \quad (5)$$

Step 2: Local feature refinement. To capture local textural patterns, the intermediate feature $\mathbf{X}'$ is passed through the L-DSCConv module. As convolutions operate on the feature dimension, we define a permutation operator $\mathcal{T}(\cdot)$ to swap the sequence and feature dimensions, mapping $\mathbb{R}^{N \times d}$ to $\mathbb{R}^{d \times N}$. The refined feature $\mathbf{X}''$ is computed as:

$$\mathbf{X}'' = \mathcal{T}^{-1}(\text{L-DSCConv}(\mathcal{T}(\text{LN}(\mathbf{X}')))) + \mathbf{X}' \quad (6)$$

Step 3: Non-linear mapping. Finally, the features are processed by the PIAF-MLP layer. To maintain the gradient signal from the attention stage, the residual connection aggregates the output of the first stage ($\mathbf{X}'$) rather than the immediate previous stage:

$$\mathbf{Y} = \text{PIAF-MLP}(\text{LN}(\mathbf{X}'')) + \mathbf{X}' \quad (7)$$

where LN denotes Layer Normalization and $\mathcal{T}^{-1}$ represents the inverse permutation. This cascaded design effectively decouples global and local feature extraction while maintaining computational efficiency.

4.2. Multi-scale depthwise separable convolutions

Standard convolutions often suffer from parameter redundancy and may be prone to overfitting on small-sample battery datasets. To mitigate this, two specialized Depthwise Separable Convolution (DSCConv) modules are designed. The kernel scales ($k = 5, k = 31$) were determined empirically to capture multi-scale degradation features, as visualized in Fig. 5.

4.2.1. S-DSCConv: Dual-role local enhancement

The small-kernel S-DSCConv employs a compact kernel size of 1×5 . It is strategically embedded within the FLFA module to fulfill a dual purpose:

- **Input Enrichment:** It captures high-frequency local variations, such as instantaneous voltage drops, to enrich the Key (**K**) and Value

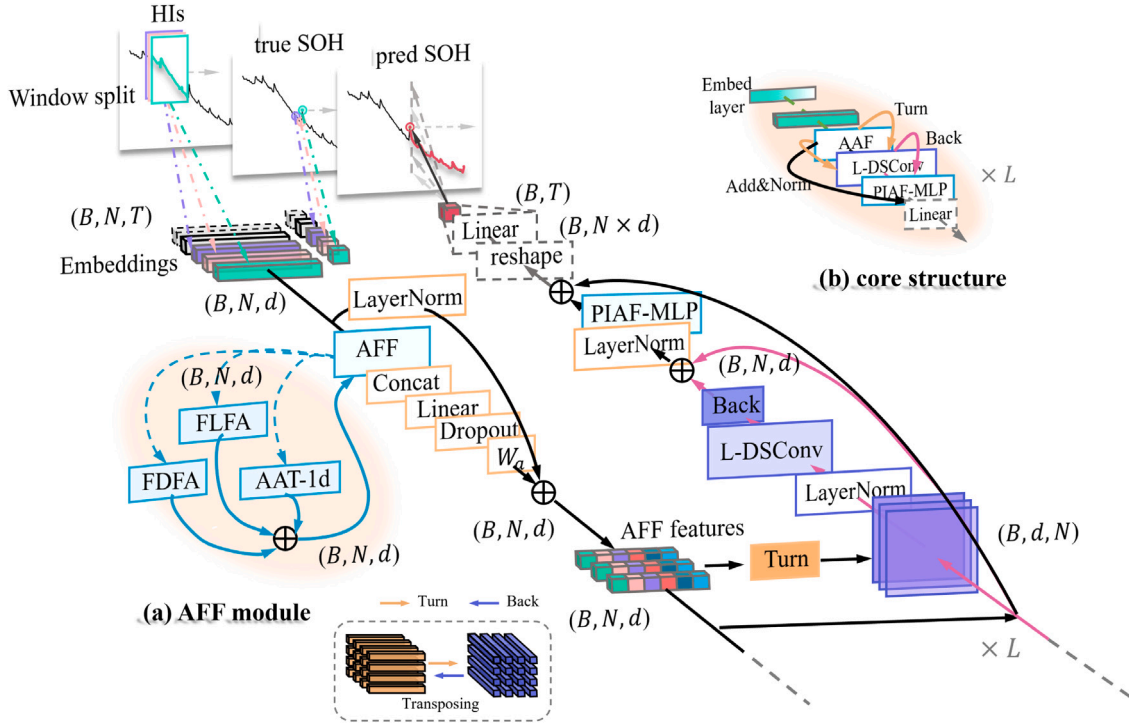


Fig. 4. Framework of SL-AgentNet. The architecture has been decomposed into modular views for clarity. (a) AFF module, (b) SL-AgentNet core structure with explicit dimension labels ($N \times d$) to illustrate the data flow.

(V) matrices. This locality bias helps mitigate feature homogenization by enabling the agent to aggregate semantically rich local patterns rather than over-smoothed noise.

- **Rank Restoration via Bypass Mechanism:** Importantly, the S-DSCov output also acts as a full-rank local branch that mitigates the attention bottleneck, as mathematically formulated in Eq. (11). While the single-agent attention ($n = 1$) inherently compresses global context into a rank-1 tensor, this parallel convolutional branch preserves the fine-grained, high-rank structural details. The fusion of these two branches addresses the low-rank bottleneck, enabling the model to simultaneously capture dominant global trends and subtle local fluctuations.

4.2.2. L-DSCov: Post-fusion global refinement

The large-kernel L-DSCov utilizes a wide kernel size of 1×31 . It functions as a standalone post-processing module following the AFF feature fusion stage.

- **Function:** With its large receptive field, this module is designed to capture long-term textural patterns and smooth out the degradation trend over extended cycles.
- **Benefit:** This module models the “aging inertia” of the battery. As demonstrated in the ablation study presented in Section 5.5, the L-DSCov plays an important role in filtering out transient sensor noise and promoting the alignment of the final SOH prediction with the dominant physical degradation trajectory rather than overfitting to local outliers.

4.3. The proposed AFF module

To circumvent the limitations of single-view feature extraction and capture diverse degradation patterns, the Attention Feature Fusion (AFF) module is proposed. Functioning as the core feature extractor, the AFF employs a tri-branch parallel architecture (Fig. 4a) to integrate orthogonal representations.

Defining the input tensor as $\mathbf{X} \in \mathbb{R}^{N \times d}$, the final output $\mathbf{Y}_{\text{AFF}}$ is derived via the element-wise summation of three distinct mechanisms:

$$\mathbf{Y}_{\text{AFF}} = \text{AAT-1d}(\mathbf{X}) + \text{FLFA}(\mathbf{X}) + \text{FDFA}(\mathbf{X}) \quad (8)$$

where FLFA($\cdot$) enhances features with local convolutional biases and FDFA($\cdot$) filters features based on amplitude significance. Of particular importance is the retention of the Agent Attention-1d (AAT-1d) branch as a linear fidelity shortcut. While the FLFA provides enhanced local context via the S-DSCov, convolutional operations impose an inductive bias that could distort pure long-range linear dependencies. Consequently, the AAT-1d branch preserves global context, promoting the retention of a stable linear reference amidst the non-linear transformations of other branches.

4.3.1. Preliminaries and the agent attention-1d (AAT-1d) branch

Standard softmax attention entails quadratic complexity $O(N^2)$, rendering it computationally challenging for long sequences. Linear attention (Han et al., 2023) addresses this limitation by utilizing a kernel feature map $\phi(\cdot)$, such as Rectified Linear Unit (ReLU). However, the direct computation of $\phi(\mathbf{K})^T \mathbf{V}$ results in a $d \times d$ matrix, incurring $O(Nd^2)$ complexity.

To achieve linear complexity $O(Nd)$ while maintaining global receptivity, the AAT-1d branch is proposed, a specialized variant of Agent Attention (Han et al., 2024) with the agent count fixed to $n = 1$. The computation is decomposed into two numerically stable steps:

- **Step 1: Global Aggregation (Complexity $O(Nd)$):** The single agent $\mathbf{A} \in \mathbb{R}^{1 \times d}$ aggregates global information from $\mathbf{K}$ and $\mathbf{V}$. To improve numerical stability against the sparsity of ReLU, given that $\phi(\cdot)$ may output zeros, a stabilization term ϵ is introduced in the normalization:

$$\mathbf{V}_{\text{agent}} = \frac{(\phi(\mathbf{A})\phi(\mathbf{K})^T) \mathbf{V}}{\phi(\mathbf{A}) \sum_k \phi(\mathbf{K})_k^T + \epsilon} \quad (9)$$

where the computation order $(\phi(\mathbf{A})\phi(\mathbf{K})^T)$ is structured to produce a $1 \times N$ vector, maintaining linear cost with respect to d .

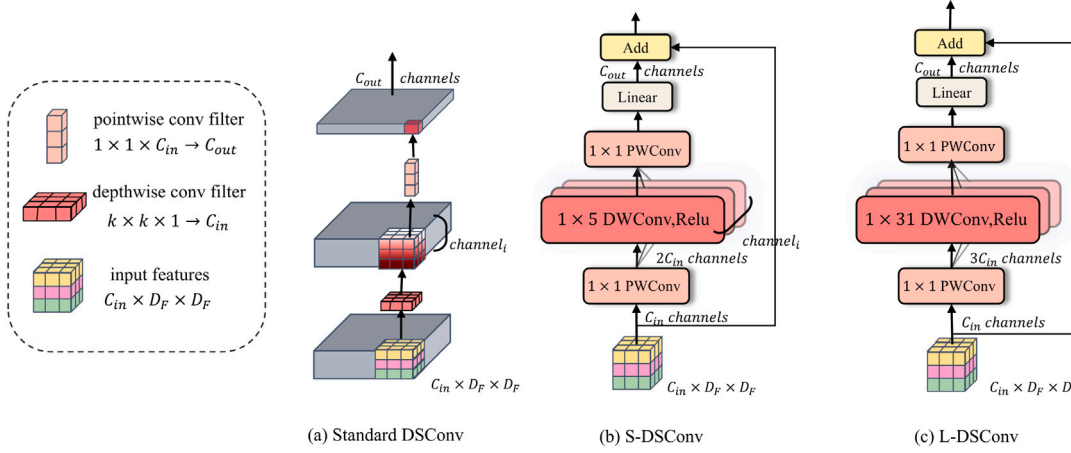


Fig. 5. Structural comparison of convolutional modules. (a) Standard DSConv. (b) The proposed S-DSConv (Kernel = 5) for local feature enhancement. (c) The proposed L-DSConv (Kernel = 31) for global trend refinement.

- **Step 2: Gated Broadcasting (Complexity $O(Nd)$):** The aggregated context $\mathbf{V}_{agent}$ is broadcast back to the queries $\mathbf{Q}$. For the single-agent case ($n = 1$), standard normalization could result in normalization collapse, a state where the query-specific term $\phi(\mathbf{Q}_i)$ is divided out, reducing the output to a static global vector for all time steps. To mitigate this loss of query selectivity, normalization is replaced with a scaling factor:

$$\text{AAT-1d}(\mathbf{X})_i = \frac{1}{\sqrt{d}} (\phi(\mathbf{Q}_i)\phi(\mathbf{A})^\top) \cdot \mathbf{V}_{agent} \quad (10)$$

This preserves the attention score $\phi(\mathbf{Q}_i)\phi(\mathbf{A})^\top$ as a dynamic gating factor, allowing each query to selectively determine the intensity of the global context it absorbs.

4.3.2. The proposed FLFA module

While the standard AAT branch captures the global baseline, it treats features as permutation-invariant tokens, potentially neglecting local sequential continuity, such as transient voltage drops. To address this deficiency and mitigate the low-rank bottleneck inherent in single-agent attention, the Fast Local-Global Fusion Attention (FLFA) module is proposed.

As depicted in Fig. 6(c), the FLFA utilizes a parallel dual-branch architecture to decouple global trend aggregation from local feature preservation.

Dual-role local enhancement (S-DSConv). The S-DSConv module (Section 4.2) is embedded into the attention mechanism to fulfill a dual purpose:

1. **Input Enrichment:** It processes the input $\mathbf{X} \in \mathbb{R}^{N \times d}$ to generate locally enhanced keys $\tilde{\mathbf{K}}$ and values $\tilde{\mathbf{V}}$. This action injects inductive bias into the attention mechanism, mitigating feature homogenization.
2. **Rank Restoration (Residual Bypass):** Importantly, the S-DSConv output $\mathbf{X}_{local}$ also serves as a full-rank residual branch. This preserves fine-grained local details that would otherwise be compressed by the low-rank global attention.

Mathematically, the projections follow an asymmetric design. The query matrix $\mathbf{Q}$ is obtained via a standard linear projection to preserve the semantic intent of the input, while the keys and values are structurally enhanced by convolution:

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_Q, \quad \mathbf{X}_{local} = \text{S-DSConv}(\mathbf{X}) \quad (11)$$

$$\tilde{\mathbf{K}} = \mathbf{X}_{local}, \quad \tilde{\mathbf{V}} = \mathbf{X}_{local} \quad (12)$$

where $\mathbf{W}_Q \in \mathbb{R}^{d \times d}$ is the query projection matrix.

Global branch (Single-agent linear attention). To capture dominant degradation trends with minimal cost, a single learnable agent $\mathbf{A} \in \mathbb{R}^{1 \times d}$ ($n = 1$) is employed. The computation follows the aggregation-broadcasting paradigm:

- **Global Aggregation:** The agent $\mathbf{A}$ aggregates context from the enhanced $\tilde{\mathbf{K}}$ and $\tilde{\mathbf{V}}$. A normalized linear formulation is utilized:

$$\mathbf{V}_{agent} = \frac{(\phi(\mathbf{A})\phi(\tilde{\mathbf{K}})^\top) \tilde{\mathbf{V}}}{\phi(\mathbf{A}) \sum_k \phi(\tilde{\mathbf{K}})_k^\top + \epsilon} \quad (13)$$

where $\phi(\cdot)$ denotes the ReLU kernel. The term $\phi(\mathbf{A})\phi(\tilde{\mathbf{K}})^\top$ computes a $1 \times N$ global attention map.

- **Gated Broadcasting:** The global context $\mathbf{V}_{agent}$ is then broadcast back to the queries. Similar to the AAT-1d branch, a scaling factor is used to prevent normalization collapse:

$$\mathbf{O}_{global} = \frac{1}{\sqrt{d}} (\phi(\mathbf{Q})\phi(\mathbf{A})^\top) \cdot \mathbf{V}_{agent} \quad (14)$$

This results in a rank-1 global context matrix $\mathbf{O}_{global} \in \mathbb{R}^{N \times d}$.

Feature fusion. Finally, to restore representational richness, the rank-1 global context is fused with the full-rank local features:

$$\mathbf{Y}_{FLFA} = \mathbf{X}_{local} + \mathbf{O}_{global} \quad (15)$$

This additive fusion integrates the aging inertia (Global Trend) with instantaneous fluctuations (Local Detail), promoting robust estimation across varying time scales while maintaining $O(Nd)$ complexity.

4.3.3. The proposed FDFA and PIAF architectures

The Fast Diversity Feature Fusion Attention (FDFA) and Parallel Informed Activation Function (PIAF) architectures were devised to mitigate potential feature collapse and enhance nonlinear representation capabilities.

FDFA: Significance filtering via orthogonal perspective. The FDFA module, illustrated in Fig. 7(b), operates as the third, specialized auxiliary branch within the AFF framework ($\mathbf{Y}_{AFF} = \text{AAT-1d}(\mathbf{X}) + \text{FLFA}(\mathbf{X}) + \text{FDFA}(\mathbf{X})$). Unlike the AAT-1d and FLFA, which primarily focus on temporal structure, the FDFA branch targets the amplitude domain, acting as a dedicated “significance filter”. Specifically, it imposes a non-linear pre-activation step:

$$\mathbf{X}_{div} = \text{ReLU}(\mathbf{X}) \quad (16)$$

This ReLU operation deliberately sparsifies the input, suppressing negative noise to isolate high-activation regions. These rectified features

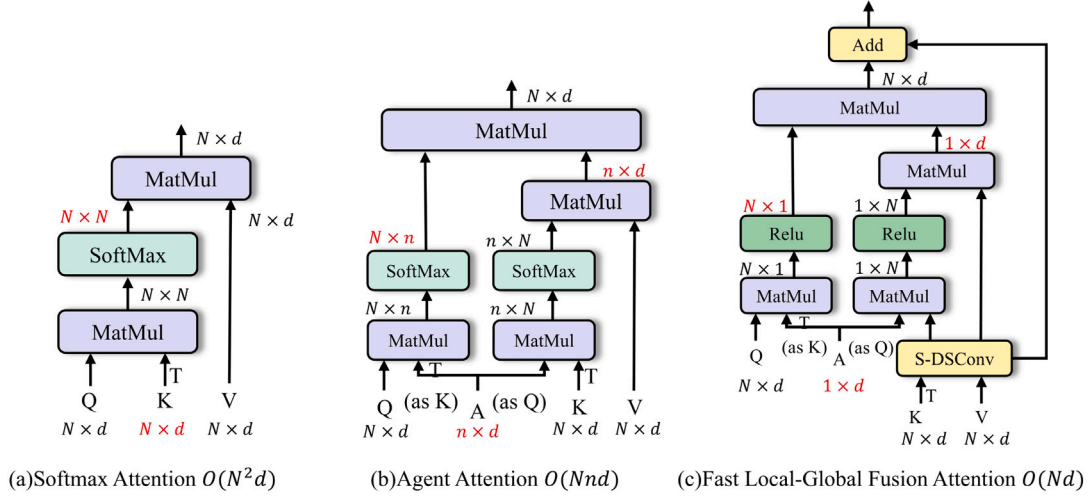


Fig. 6. Structural representation of different models: (a) Softmax Attention ($O(N^2)$), (b) Agent Attention ($O(N)$), and (c) the proposed Fast Local-Global Fusion Attention (FLFA), which integrates S-DSCnv for local enhancement and rank restoration.

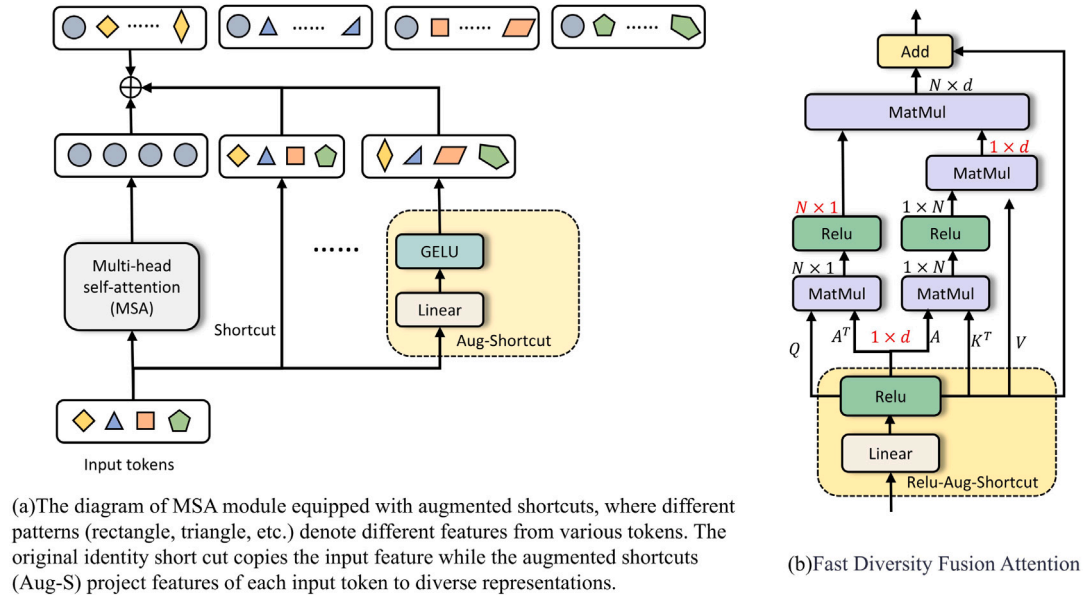


Fig. 7. Schematic comparison between (a) a traditional Multi-Head Self-Attention (MSA) module equipped with Aug-Shortcuts and (b) the proposed Fast Diversity Fusion Attention (FDFA).

subsequently traverse a standard Agent Attention block, preserving linear complexity $O(Nd)$:

$$\mathbf{Y}_{\text{FDFA}} = \text{AAT-1d}(\mathbf{X}_{\text{div}}) \quad (17)$$

By fusing this output, the model assimilates both the raw global context and salient high-energy patterns from an orthogonal perspective. This strategy facilitates effective fusion by providing a complementary, sparsity-controlled feature subspace to the final AFF output.

PIAF: Parallel heterogeneous activation fusion. The PIAF module, depicted in Fig. 8(b), serves as the final non-linear mapping stage, designed to address complex degradation patterns that homogeneous activation functions struggle with. In contrast to architectures relying on single activation types, PIAF leverages a “Mixture of Activations” paradigm to engineer a heterogeneous feature space by combining three mathematically distinct and physically motivated branches in parallel:

- **Branch 1 (Sparsity via ReLU):** This branch isolates sparse, high-energy features, effectively capturing dominant degradation events and introducing important non-linearity.
- **Branch 2 (Bipolarity via Tanh):** This branch offers a bounded non-linearity within the range $(-1, 1)$. It is essential for modeling saturation dynamics, such as the voltage plateaus and capacity regeneration limits frequently observed in electrochemical data, exemplified by the NASA dataset.
- **Branch 3 (Linearity via Identity):** This branch maintains the intrinsic linear geometry of input features. It facilitates linear fidelity and promotes robust gradient flow, operating as a shortcut connection consistent with the lightweight philosophy (He et al., 2015a).

Formalizing this structure, the branch operations are defined as:

$$\mathbf{H}_{\text{sparse}} = \text{ReLU}(\mathbf{X}\mathbf{W}_1), \quad \mathbf{H}_{\text{bipolar}} = \text{Tanh}(\mathbf{X}\mathbf{W}_2), \quad \mathbf{H}_{\text{linear}} = \mathbf{X}\mathbf{W}_3 \quad (18)$$

The final output aggregates these diverse representations:

$$\mathbf{Y}_{\text{PIAF}} = \mathbf{H}_{\text{sparse}} + \mathbf{H}_{\text{bipolar}} + \mathbf{H}_{\text{linear}} \quad (19)$$

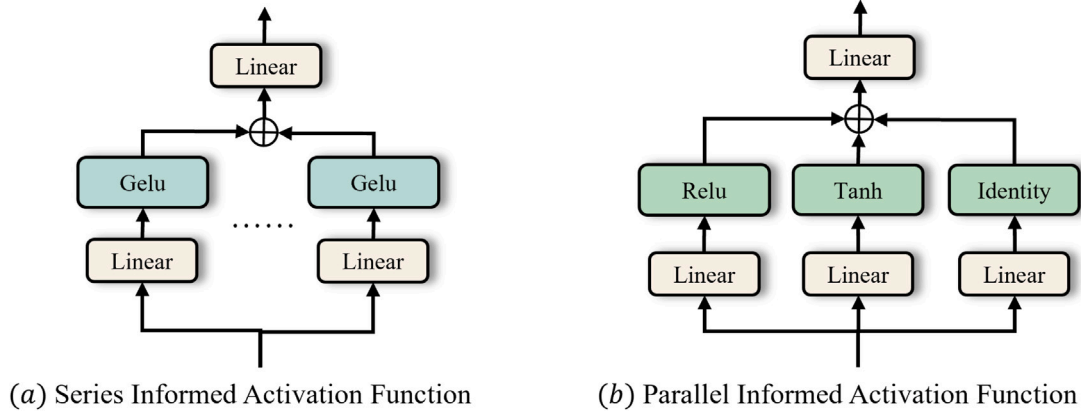


Fig. 8. Comparison of (a) a traditional Series Informed Activation Function and (b) the proposed Parallel Informed Activation Function (PIAF).

where $W_1, W_2, W_3 \in \mathbb{R}^{d \times d}$ denote learnable weight matrices. By synergizing these complementary mathematical properties, PIAF yields a feature richness enhanced compared to that of homogeneous activation layers.

5. Results and discussion

This section presents a systematic evaluation of the proposed SL-AgentNet framework, benchmarked against four baseline models, including the Transformer, LSTM, CNN-Transformer, and CNN-LSTM, across three authoritative battery datasets. To ensure experimental rigor and simulate a realistic “early prediction” scenario, we adhered to a strict protocol where models were trained using only the initial 30% of cycle data. Consequently, the remaining 70% served as the unseen test horizon for prognostic evaluation.

In addition to reporting estimation accuracy, this section elucidates the mechanistic origins of performance gains, verifies model robustness through rigorous ablation and sensitivity analyses, quantifies prediction uncertainty, and addresses the limitations of the current approach. This holistic assessment validates the practical viability of the framework for resource-constrained vehicular instrumentation.

5.1. Evaluation metrics

To comprehensively assess the performance of the proposed framework, we employed a multi-dimensional evaluation protocol comprising both deterministic error metrics and probabilistic uncertainty metrics.

5.1.1. Deterministic accuracy metrics

Four standard metrics were utilized to gauge estimation precision (Chicco et al., 2021): Coefficient of Determination (R^2), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Root Mean Square Error (RMSE). While MAE and MAPE reflect general accuracy, RMSE is particularly critical for BMS safety, as it penalizes larger prediction errors that could lead to hazardous overcharging or over-discharging. These metrics are defined as:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad \text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (20)$$

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|, \quad R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (21)$$

where y_i and $\hat{y}_i$ represent the ground-truth and predicted SOH values, respectively; $\bar{y}$ denotes the mean of observed values, and n is the sample size.

5.1.2. Uncertainty quantification metrics

To evaluate the reliability of the prediction intervals (PIs), we adopted three probabilistic metrics based on established calibration frameworks (Kuleshov et al., 2018):

- **Prediction Interval Coverage Probability (PICP):** This metric measures the proportion of ground-truth values falling within the predicted intervals $[L_i, U_i]$ at a nominal confidence level:

$$\text{PICP} = \frac{1}{n} \sum_{i=1}^n \mathbb{I}\{y_i \in [L_i, U_i]\} \quad (22)$$

where $\mathbb{I}\{\cdot\}$ is the indicator function.

- **Mean Prediction Interval Width (MPIW):** This metric quantifies the sharpness, or efficiency, of the uncertainty bounds:

$$\text{MPIW} = \frac{1}{n} \sum_{i=1}^n (U_i - L_i) \quad (23)$$

- **Average Calibration Error (ACE):** To summarize global calibration quality, we report the ACE computed as the average absolute deviation across a discretized grid C of nominal confidence levels (e.g., $C = \{0.5, 0.8, 0.9, 0.95\}$) (Kuleshov et al., 2018):

$$\text{ACE} = \frac{1}{|C|} \sum_{c \in C} |\text{PICP}(c) - c| \quad (24)$$

This metric quantifies the average misalignment between the expected coverage c and the observed coverage $\text{PICP}(c)$, where a value closer to 0 indicates superior calibration.

5.2. Model robustness and hyperparameter analysis

To address theoretical concerns regarding training stability and validate the robustness of our lightweight design choices, the initialization strategy, convergence dynamics, and hyperparameter sensitivity were analyzed.

5.2.1. Initialization strategy and convergence analysis

Unbounded kernel functions, such as ReLU, often pose challenges to the training stability of linear attention mechanisms. To promote robust convergence for SL-AgentNet, its initialization strategy and convergence behavior were analyzed using quantitative metrics derived from the training logs of representative battery cells (NASA B0005 and CALCE CS2_35).

Initialization strategy. By fixing the number of agent tokens to $n = 1$, the theoretical lower bound linear complexity $O(Nd)$ was achieved. In contrast to standard Agent Attention implementations that derive agents via dynamic pooling, the single agent token $\mathbf{A} \in \mathbb{R}^{1 \times d}$ was defined as a static learnable anchor. While this design provides a

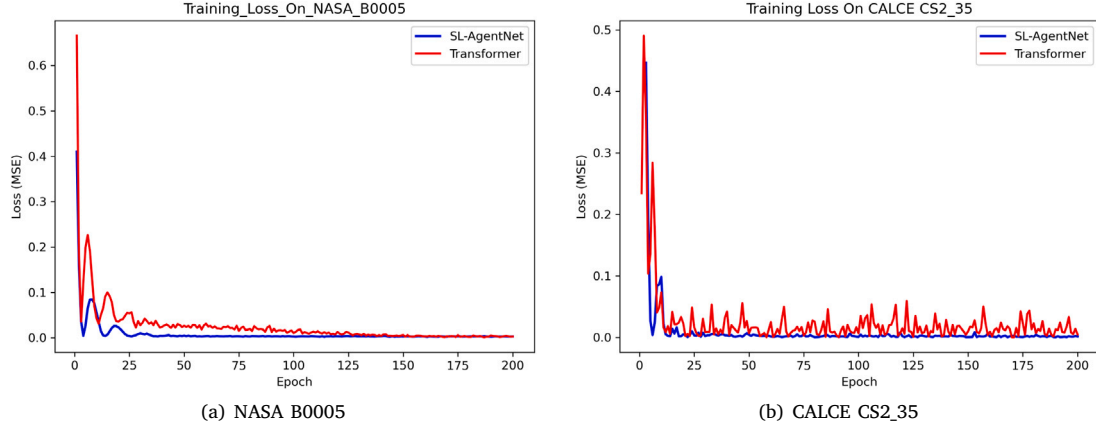


Fig. 9. Comparison of training loss convergence dynamics. (a) On NASA B0005, SL-AgentNet achieves rapid convergence, crossing the 10% relative loss threshold at epoch 4, compared to epoch 58 for the Transformer. (b) On CALCE CS2_35, SL-AgentNet maintains a significantly lower and smoother loss profile in the late training phase, validating the stabilizing effect of the proposed initialization and filtering mechanisms.

consistent global reference, it introduces a potential stability risk: if initialized with large negative magnitudes, $\mathbf{A}$ may cause specific feature dimensions to become permanently deactivated by the ReLU kernel, a phenomenon known as the dead neuron problem (He et al., 2015b). To mitigate this risk, a Truncated Normal Distribution strategy was adopted for $\mathbf{A}$:

$$\mathbf{A} \sim \mathcal{N}_{trunc}(\mu = 0, \sigma = 0.02, a = -2\sigma, b = 2\sigma) \quad (25)$$

$\mu = 0$ was selected rather than a positive bias to maintain variance consistency with standard initialization schemes such as Xavier (Glorot and Bengio, 2010), thereby avoiding distributional shifts that could destabilize downstream layers. The truncation aims to mitigate extreme negative outliers; by confining initial values near zero, the aim was to maintain even negative entries within a recoverable range, allowing gradients to reactivate them during early training phases.

Convergence properties. The training dynamics of SL-AgentNet were benchmarked against the standard Transformer baseline. To quantify the convergence speed, a relative convergence threshold was defined, marking the epoch where the training loss first drops below 10% of the initial mean loss. As illustrated in Fig. 9, quantitative analysis reveals two distinct advantages:

- **Rapid Optimization (NASA B0005):** SL-AgentNet demonstrates improved optimization efficiency on the NASA B0005 cell. Under the defined 10% threshold criterion, the model enters the low-error regime within just 4 epochs (threshold ≈ 0.010), whereas the Transformer requires 58 epochs (threshold ≈ 0.018). Furthermore, the Cumulative Training Loss (CTL), calculated as the integral of the loss curve over time, is approximately 2.8 times lower for SL-AgentNet than for the Transformer (1.83 vs. 5.14), indicating a much more direct optimization trajectory.
- **Stability and Smoothness (CALCE CS2_35):** In the asymptotic convergence phase (the final 20 epochs) on the CALCE CS2_35 cell, SL-AgentNet exhibits enhanced stability. The standard deviation of the loss is reduced by an order of magnitude compared to the baseline (9.4×10^{-4} vs. 8.1×10^{-3}). This stability may be attributed to the S-DConv module embedded in the FLFA, which acts as a local smoothing mechanism. This structural prior implicitly suppresses high-frequency noise in the gradients, reducing the oscillation often observed in unnormalized linear attention architectures.

5.2.2. Hyperparameter sensitivity and selection methodology

To optimize the model architecture while adhering to the “Early Prediction” constraint, a historical reference strategy was adopted. This

Table 6

Hyperparameter grid search space.

Hyperparameter	Search space/Values
Learning rate	[0.001, 0.01]
Network depth (L)	[1, 2, 4, 8]
Dense dimension	[16, 32, 64, 128]
Embedding dimension (d)	[16, 32, 64, 128]

approach simulates a realistic deployment scenario where hyperparameters are calibrated offline using historical run-to-failure data prior to deployment on active vehicles.

Calibration protocol. For each dataset, the first battery (e.g., NASA B0005) was designated as the historical reference cell. The hyperparameter selection process followed a protocol to minimize methodological data leakage:

1. **Training Split:** The model was trained using only the first 30% of the reference cell’s data.
2. **Validation Split:** The performance of distinct hyperparameter configurations was evaluated on the remaining 70% of the reference cell. This tests the model’s ability to extrapolate from early-life patterns to end-of-life degradation.
3. **Fixed Deployment:** The optimal configuration identified from this historical reference was subsequently frozen and applied to all other test batteries (e.g., B0006, B0007, B0018) without further tuning.

The grid search covered key structural parameters as detailed in Table 6. Fig. 10 visualizes the sensitivity of R^2 on the NASA B0005 reference cell during the validation phase.

Sensitivity to the embedding dimension (d) and network depth (L) was specifically analyzed. The analysis reveals consistent trends:

- **Optimal Configuration (Fig. 10):** The configuration ($d = 16, L = 1$) achieved the highest validation accuracy ($R^2 = 0.9866$). This indicates that a compact feature space is sufficient to capture the degradation dynamics.
- **Validity of Shallow Attention ($L = 1$):** Increasing the attention depth to $L = 4$ resulted in a performance drop ($R^2 \approx 0.94$), a result consistent with the principle of parsimony. Although $L = 1$ may appear shallow, the SL-AgentNet is not representationally limited. The integrated S-DConv module provides local feature extraction, and the multi-branch PIAF module contributes significant non-linear depth. Consequently, a single global attention block is

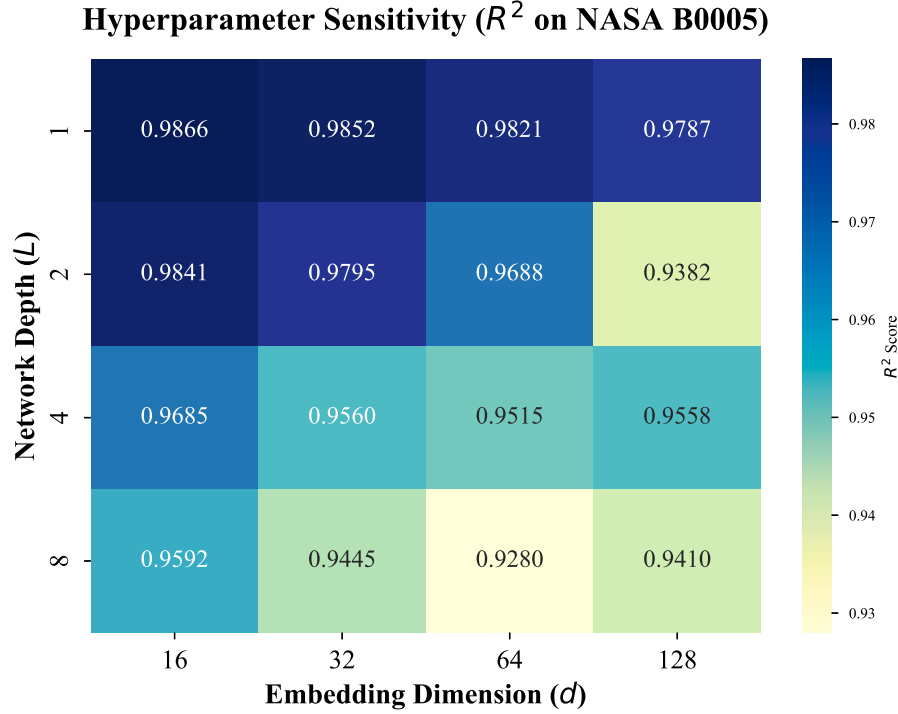


Fig. 10. Hyperparameter sensitivity analysis (R^2) performed on the NASA B0005 historical reference cell. The heatmap shows validation performance on the unseen latter 70% of the cycle life (model trained on first 30%). Darker blue indicates higher accuracy. The selected optimal configuration is ($d = 16, L = 1$).

sufficient to aggregate these rich features without the overfitting risks associated with deeper stacks on sparse battery data.

Based on these validation results on the historical reference, the robust configuration of $d = 16, L = 1$ was selected for all subsequent testing.

5.3. Comparison on Oxford dataset: Handling abrupt degradation

Fig. 11 and Table 7 present the estimation results on the Oxford dataset. While all models capture the general linear decay trend, a granular analysis of the degradation trajectories reveals critical differences in dynamic response characteristics.

5.3.1. Responsiveness to instantaneous drops (Cells 2 & 6)

A challenge inherent to the Oxford dataset is the presence of abrupt SOH drops or local knee-points, which are particularly visible in Cell 2 and Cell 6.

- **Baseline Limitations (Hysteresis Effect):** As evidenced by the magnified regions of Fig. 11, sequential models such as the LSTM and CNN-LSTM often suffer from a pronounced hysteresis effect, or phase lag. Constrained by recurrent memory inertia, these architectures may struggle to synchronize with rapid downward transitions. This results in a negative time lag and a consistent underestimation of the instantaneous degradation magnitude.
- **SL-AgentNet Advantage:** In contrast, SL-AgentNet exhibits improved temporal alignment, closely tracking these abrupt drops with minimal lag. Mechanistically, this responsiveness is driven by the S-DConv module embedded within the FLFA block. By executing convolution operations prior to attention aggregation, the S-DConv acts as a high-frequency feature extractor. This component remains highly sensitive to immediate gradient changes, allowing the model to adapt rapidly to local structural shifts without the memory inertia typical of RNNs.

5.3.2. Precision in linear decay (Cells 1, 3, 7, & 8)

For cells exhibiting relatively smooth linear decay, such as Cells 1, 3, 7, and 8, the primary challenge involves minimizing steady-state error and jitter caused by measurement noise. SL-AgentNet achieves strong alignment with the ground truth, yielding R^2 values exceeding 0.998; for instance, Cell 3 achieves 0.9990. Here, the post-fusion L-DConv module plays an important role. Its large receptive field smooths out sensor noise while preserving the dominant aging trend. Consequently, SL-AgentNet achieves a lower average MAE of 0.0029 and an RMSE of 0.0041, representing a 60.8% reduction in MAE compared to the Transformer baseline (0.0074). This demonstrates its capability to balance dynamic sensitivity via the S-DConv with static stability via the L-DConv.

5.4. Generalization analysis: Capacity regeneration and non-linearity

The NASA and CALCE datasets present more complex aging dynamics, including significant capacity regeneration and accelerated non-linear tails. The robustness of SL-AgentNet is analyzed as follows (refer to Table 8 and Fig. 12):

Capturing capacity regeneration (NASA case). The NASA batteries, exemplified by B0005, are characterized by frequent capacity regeneration phenomena that appear as local upward spikes in the SOH curve.

- **Observation:** Standard Transformers often overreact to these local variations or treat them as noise due to a lack of structural regularization. In contrast, SL-AgentNet captures both the amplitude and timing of these regeneration peaks, achieving a low RMSE of 0.0072 on B0005.
- **Mechanism:** This fine-grained capture capability is attributed to the FDFA module. By introducing a non-linear pre-activation (ReLU) branch, the FDFA acts as a significance filter that highlights high-energy local changes, such as regeneration spikes,

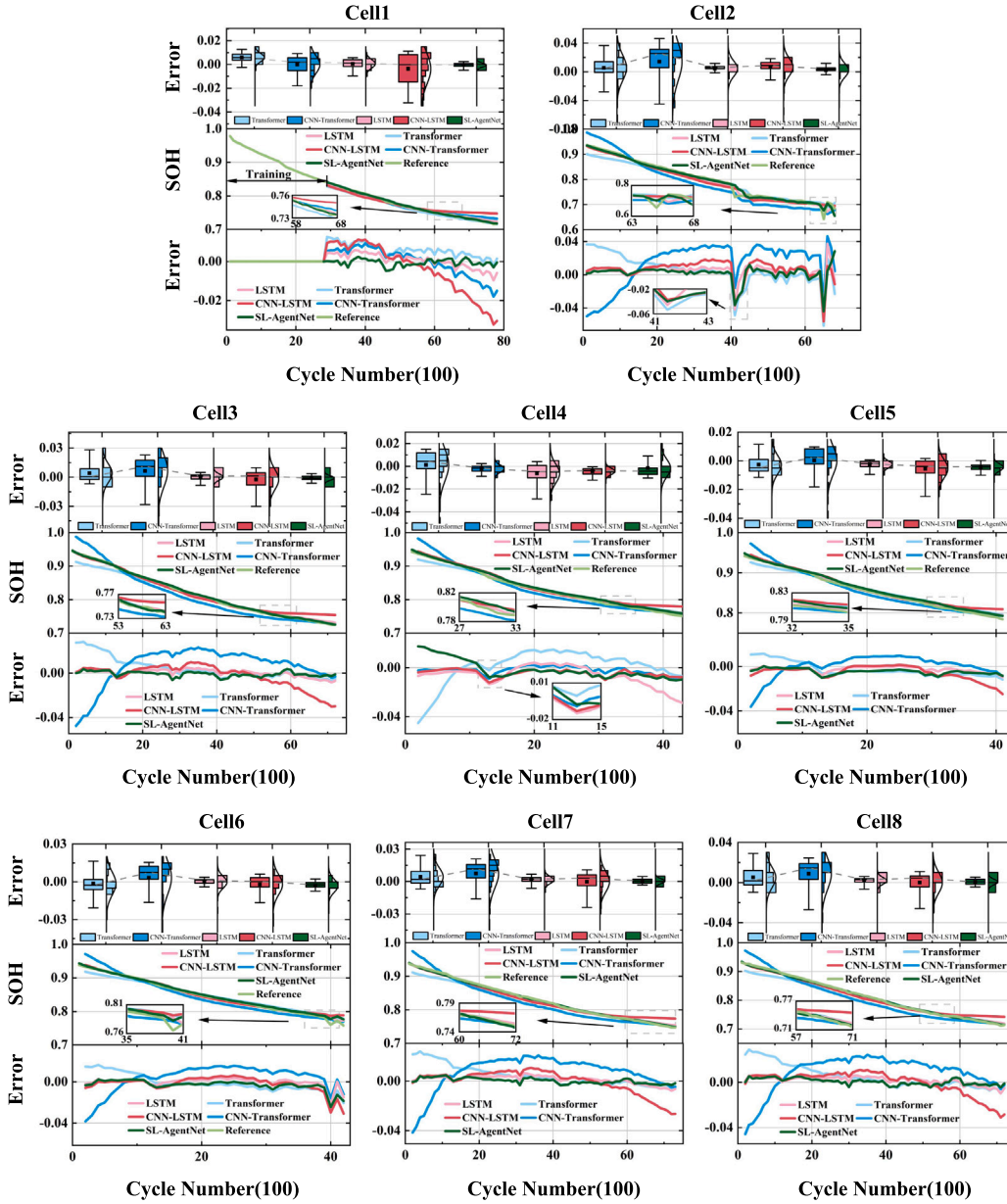


Fig. 11. The experimental results of different models on the Oxford dataset.

against the background of gradual decay. This enables the model to distinguish meaningful physical recovery from random measurement noise.

Tracking accelerated aging trends (CALCE case). The CALCE batteries, such as CS2_38, exhibit a rapid “diving” behavior or accelerated aging in the late life cycle.

- **Observation:** While the response of SL-AgentNet to minute instantaneous drops in this dataset is moderate compared to the NASA case, its overall fitting precision regarding the accelerated decay trajectory is significantly better than the baselines. For instance, on CS2_38, SL-AgentNet achieves an RMSE of 0.0249, outperforming both the Transformer (0.0344) and LSTM (0.0331).
- **Mechanism:** This robustness is attributed to the Global Agent Anchor. Unlike standard attention mechanisms which compute pairwise $N \times N$ dependencies and often dilute the trend with

local noise, our FLFA module aggregates information into a single static agent. This agent learns the global degradation prototype, prioritizing the dominant accelerated aging trend over transient fluctuations. This contributes to the model maintaining a tight global fit even during highly non-linear late-life stages.

In summary, SL-AgentNet achieves an average RMSE of 0.0169 across the combined datasets, outperforming the Transformer (0.0402) and LSTM (0.0260). This demonstrates that our “Local-Global” architectural design generalizes effectively across different battery chemistries and degradation modes.

5.5. Ablation study and contribution decoupling

To evaluate the contributions of the proposed framework and disentangle the sources of performance gains, a systematic ablation study

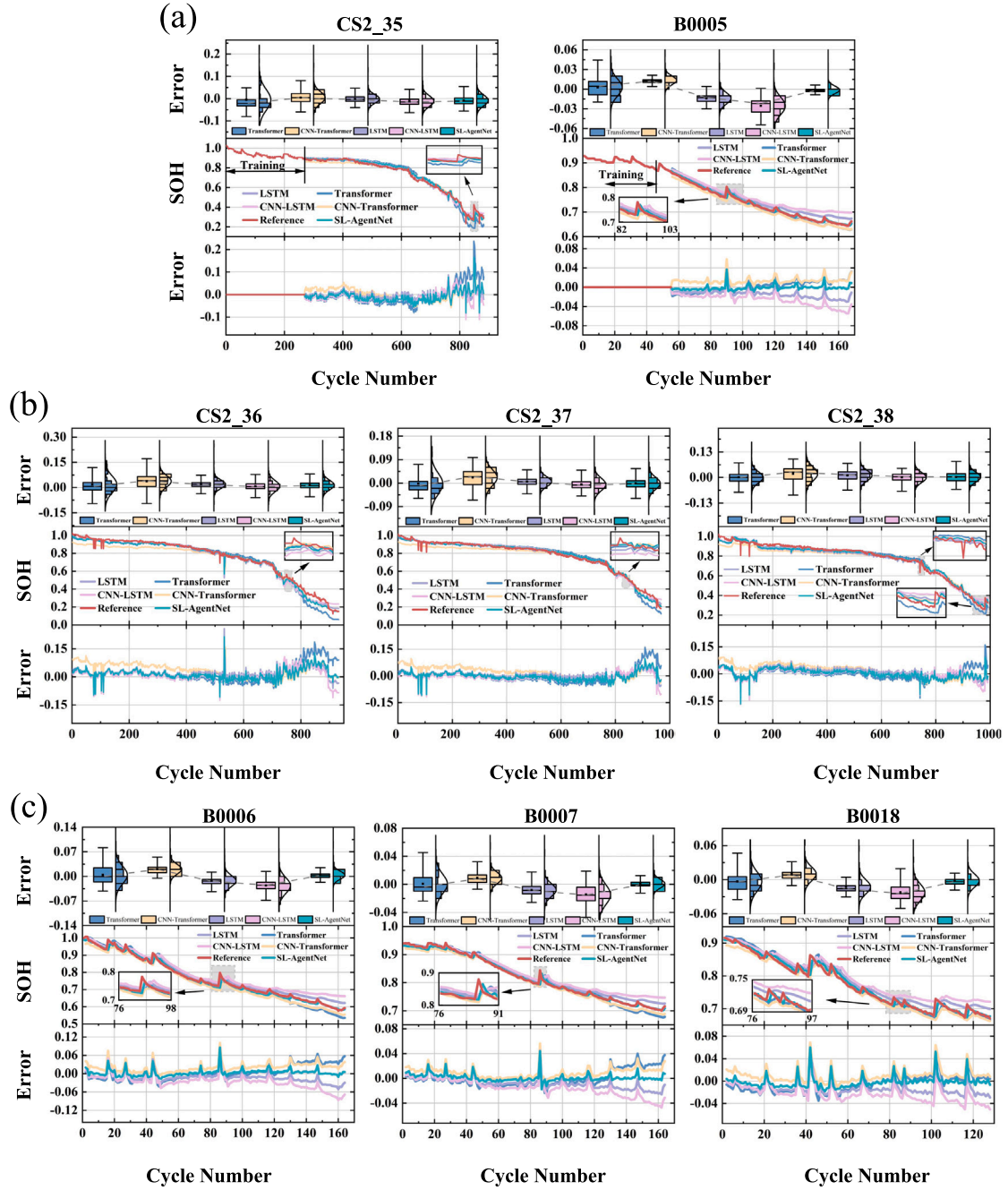


Fig. 12. (a) The validation results of five models on CS2_35 and B0005. (b) The test results using five models trained on CS2_35. (c) The test results using five models trained on B0005.

was conducted. This analysis is structured into three tiers: a macro-level decoupling of Health Indicators (HIs) versus model architecture, a micro-level analysis of specific component effectiveness, and an interaction analysis of the attention mechanisms.

5.5.1. Macro-ablation: Decoupling HIs and model architecture

A pivotal research inquiry concerns the attribution of performance gains: do they arise from the proposed Systematic HIs (CCCA/CCDA) or the specialized SL-AgentNet architecture? To decouple these contributions and position our work within the landscape of state-of-the-art methods, our approach was benchmarked against recent results reported in the literature on identical datasets.

Experimental Constraint (The “Early Prediction” Challenge): Importantly, to demonstrate data efficiency, an “Early Prediction” protocol was adopted where our model was trained using only the first 30% of cycle data. This stands in contrast to benchmark methods, which typically utilize 50%–70% of lifecycle data, thereby possessing an informational advantage.

As presented in Table 9, the comparison reveals two key insights:

1. *Contribution of Systematic HIs (High Information Density):* Standard LSTM models employing traditional manual/IC features and trained on 50% data achieve an RMSE of 0.0183, as reported by Yan et al. (2025). Importantly, when employing our

Table 7

SOH estimation errors of different models on the Oxford dataset.

Cells	Methods	R^2	MAE	MAPE	RMSE
Cell1*	SL-AgentNet	0.9985	0.0011	0.0015	0.0014
	Transformer	0.9649	0.0051	0.0045	0.0077
	LSTM	0.9895	0.0028	0.0023	0.0041
	CNN-Transformer	0.9616	0.0053	0.0044	0.0079
	CNN-LSTM	0.8687	0.0098	0.0077	0.0139
Cell2	SL-AgentNet	0.9809	0.0063	0.0084	0.0109
	Transformer	0.9444	0.0136	0.0096	0.0159
	LSTM	0.9779	0.0085	0.0058	0.0103
	CNN-Transformer	0.8576	0.0217	0.0204	0.0348
	CNN-LSTM	0.9686	0.0102	0.0081	0.0142
Cell3	SL-AgentNet	0.9990	0.0016	0.0019	0.0021
	Transformer	0.9752	0.0076	0.0051	0.0079
	LSTM	0.9977	0.0023	0.0019	0.0032
	CNN-Transformer	0.9270	0.0131	0.0110	0.0177
	CNN-LSTM	0.9738	0.0078	0.0056	0.0098
Cell4	SL-AgentNet	0.9932	0.0040	0.0048	0.0047
	Transformer	0.9801	0.0058	0.0050	0.0081
	LSTM	0.9954	0.0028	0.0021	0.0035
	CNN-Transformer	0.9373	0.0103	0.0075	0.0118
	CNN-LSTM	0.9648	0.0077	0.0057	0.0096
Cell5	SL-AgentNet	0.9903	0.0042	0.0049	0.0046
	Transformer	0.9760	0.0052	0.0049	0.0078
	LSTM	0.9944	0.0025	0.0019	0.0030
	CNN-Transformer	0.9530	0.0073	0.0051	0.0079
	CNN-LSTM	0.9659	0.0062	0.0044	0.0072
Cell6	SL-AgentNet	0.9910	0.0030	0.0036	0.0049
	Transformer	0.9665	0.0068	0.0053	0.0086
	LSTM	0.9914	0.0034	0.0019	0.0032
	CNN-Transformer	0.9280	0.0099	0.0082	0.0131
	CNN-LSTM	0.9734	0.0060	0.0036	0.0061
Cell7	SL-AgentNet	0.9990	0.0015	0.0018	0.0018
	Transformer	0.9723	0.0069	0.0049	0.0075
	LSTM	0.9961	0.0026	0.0022	0.0036
	CNN-Transformer	0.9258	0.0112	0.0096	0.0154
	CNN-LSTM	0.9709	0.0070	0.0054	0.0091
Cell8	SL-AgentNet	0.9989	0.0017	0.0021	0.0022
	Transformer	0.9707	0.0084	0.0059	0.0093
	LSTM	0.9967	0.0028	0.0025	0.0041
	CNN-Transformer	0.9274	0.0132	0.0114	0.0187
	CNN-LSTM	0.9751	0.0077	0.0060	0.0104
Average	SL-AgentNet	0.9938	0.0029	0.0036	0.0041
	Transformer	0.9688	0.0074	0.0057	0.0091
	LSTM	0.9923	0.0035	0.0026	0.0044
	CNN-Transformer	0.9272	0.0115	0.0097	0.0159
	CNN-LSTM	0.9577	0.0078	0.0058	0.0100

Cell1* indicates that the model was trained on this cell using 30% of the early data, and its values are averaged among ten experiments. The best results are marked in bold and underlined.

proposed Systematic HIs (CCCA/CCDA) with a standard LSTM, even under the strict limitation of 30% training data, the model maintains a comparable RMSE of 0.0211. Moreover, equipping a standard Transformer with our HIs yields an RMSE of 0.0150, matching or marginally outperforming the baseline reported by Xu et al. (2024) (RMSE 0.0151), despite the data scarcity. These results suggest that our screened HIs possess high “information density”, enabling standard models to extract robust degradation patterns from sparse early-life data where traditional features often fail.

2. *Contribution of SL-AgentNet Architecture (Architectural Efficacy):* A notable performance enhancement is attributed to the model architecture. Under the identical 30% data constraint, upgrading from a standard Transformer (RMSE 0.0150) to the proposed SL-AgentNet (RMSE 0.0052) yields a 65.3% error reduction. This suggests that while Systematic HIs provide a robust, noise-immune foundation, the architectural innovations, specifically the linear-complexity FLFA, the multi-branch PIAF, and the S-DSCConv, play a key role in modeling the complex, non-linear dependencies required for high-precision early prediction.

Table 8

Estimation results on the CALCE and NASA datasets.

Cell	Methods	R^2	MAE	MAPE	RMSE
B0005*	SL-AgentNet	0.9866	0.0050	0.0068	0.0072
	Transformer	0.9553	0.0130	0.0109	0.0150
	LSTM	0.9280	0.0165	0.0149	0.0211
	CNN-Transformer	0.9402	0.0151	0.0135	0.0187
	CNN-LSTM	0.7802	0.0289	0.0263	0.0374
B0006	SL-AgentNet	0.9885	0.0086	0.0111	0.0130
	Transformer	0.9564	0.0252	0.0199	0.0274
	LSTM	0.9713	0.0204	0.0172	0.0238
	CNN-Transformer	0.9568	0.0251	0.0207	0.0285
	CNN-LSTM	0.9191	0.0343	0.0294	0.0418
B0007	SL-AgentNet	0.9924	0.0045	0.0054	0.0069
	Transformer	0.9699	0.0136	0.0108	0.0139
	LSTM	0.9766	0.0120	0.0100	0.0128
	CNN-Transformer	0.9762	0.0121	0.0094	0.0116
	CNN-LSTM	0.9337	0.0203	0.0166	0.0217
B0018	SL-AgentNet	0.9845	0.0084	0.0108	0.0121
	Transformer	0.9459	0.0170	0.0133	0.0169
	LSTM	0.9422	0.0176	0.0160	0.0212
	CNN-Transformer	0.9523	0.0160	0.0111	0.0144
	CNN-LSTM	0.8668	0.0267	0.0238	0.0321
CS2_35*	SL-AgentNet	0.9867	0.0148	0.0251	0.0218
	Transformer	0.9380	0.0472	0.0363	0.0702
	LSTM	0.9895	0.0194	0.0132	0.0232
	CNN-Transformer	0.9797	0.0270	0.0207	0.0333
	CNN-LSTM	0.9800	0.0268	0.0214	0.0372
CS2_36	SL-AgentNet	0.9834	0.0193	0.0367	0.0303
	Transformer	0.9491	0.0531	0.0355	0.0901
	LSTM	0.9814	0.0321	0.0247	0.0475
	CNN-Transformer	0.9533	0.0509	0.0424	0.0607
	CNN-LSTM	0.9798	0.0334	0.0228	0.0551
CS2_37	SL-AgentNet	0.9902	0.0131	0.0232	0.0187
	Transformer	0.9541	0.0404	0.0278	0.0535
	LSTM	0.9895	0.0194	0.0138	0.0249
	CNN-Transformer	0.9621	0.0367	0.0313	0.0438
	CNN-LSTM	0.9833	0.0244	0.0178	0.0329
CS2_38	SL-AgentNet	0.9819	0.0205	0.0303	0.0249
	Transformer	0.9773	0.0279	0.0207	0.0344
	LSTM	0.9780	0.0275	0.0229	0.0331
	CNN-Transformer	0.9535	0.0399	0.0340	0.0457
	CNN-LSTM	0.9813	0.0253	0.0187	0.0306
Average	SL-AgentNet	0.9868	0.0118	0.0187	0.0169
	Transformer	0.9558	0.0297	0.0219	0.0402
	LSTM	0.9696	0.0206	0.0166	0.0260
	CNN-Transformer	0.9592	0.0278	0.0229	0.0321
	CNN-LSTM	0.9281	0.0275	0.0221	0.0361

B0005* and CS2_35* denote that they are the training sets with 30% of the early data. The best results are marked in bold and underlined.

5.5.2. Micro-ablation I: Component effectiveness analysis

To validate the efficacy of key architectural components, ablation studies were conducted focusing on convolutional modules and activation functions across the Oxford, NASA, and CALCE datasets.

Impact of depthwise separable convolutions (S/L-DSCConv). The proposed framework integrates two distinct convolutional modules: the small-kernel S-DSCConv embedded within the FLFA module for pre-attention local enhancement, and the large-kernel L-DSCConv positioned after the AFF module for post-fusion refinement. Their specific contributions were assessed by substituting them with standard convolutions. Although standard convolutions possess higher parameter counts and theoretically larger capacities, the results in Table 10 indicate performance degradation, thereby supporting the efficiency of our design:

- **Pre-Attention Local Enhancement (S-DSCConv):** Substituting the S-DSCConv with a standard convolution corresponds to an accuracy drop, notably increasing the RMSE from 0.0041 to 0.0102 on the Oxford dataset. This suggests that the parameter-heavy standard convolution may be prone to overfitting on sparse battery data.

Table 9

Macro-Ablation: Decoupling contributions against literature benchmarks (NASA B0005).

Source	Methodology (Model + Feature)	Train data	RMSE	MAE
<i>Literature Benchmarks (Traditional HIs)</i>				
Yan et al. Yan et al. (2025)	Standard LSTM + 12 Manual Feat.	50%	0.0183	0.0160
Xu et al. Xu et al. (2024)	Transformer + IC Peak Area	Start Cyc 70 ^a	0.0151	0.0105
Huang et al. Huang et al. (2025)	Swin Transformer + Multi-features	50%	0.0097	0.0072
<i>Our Framework (Systematic HIs: CCCA/CCDA)</i>				
Our baseline	Standard Transformer	30%	0.0150	0.0130
Our baseline	Standard LSTM	30%	0.0211	0.0165
Proposed	SL-AgentNet	30%	0.0052	0.0062

^a Xu et al. utilize a fixed starting point (cycle 70) for prediction.**Table 10**

Ablation study of convolutional modules (average results).

Dataset	Convolution configuration	R ²	RMSE	MAE
NASA	Std. Conv (Replaces S-DConv)	0.9764	0.0121	0.0099
	Std. Conv (Replaces L-DConv)	0.8933	0.0259	0.0237
	Full SL-AgentNet (S+L DConv)	0.9880	0.0098	0.0066
Oxford	Std. Conv (Replaces S-DConv)	0.9627	0.0102	0.0075
	Std. Conv (Replaces L-DConv)	0.9720	0.0087	0.0071
	Full SL-AgentNet (S+L DConv)	0.9938	0.0041	0.0029
CALCE	Std. Conv (Replaces S-DConv)	0.8422	0.0686	0.0512
	Std. Conv (Replaces L-DConv)	0.8240	0.0719	0.0547
	Full SL-AgentNet (S+L DConv)	0.9856	0.0239	0.0169

In contrast, the S-DConv functions as a structural regularizer; by decoupling spatial and channel correlations, it imposes a beneficial inductive bias that encourages the model to focus on essential high-frequency local patterns rather than noise.

- **Post-Fusion Global Refinement (L-DConv):** Replacing the L-DConv leads to a performance decline, particularly on the CALCE dataset where the RMSE increases to 0.0719. This suggests that a lightweight, large-kernel convolution is beneficial for smoothing fused attention features and extracting long-term degradation trends prior to the final regression stage.

Impact of heterogeneous activation (PIAF vs. Standard MLP). The PIAF module functions as the final regression head. This heterogeneous design, which combines ReLU, Tanh, and Identity functions, was compared against a standard MLP baseline configured with homogeneous Gaussian Error Linear Unit (GeLU) activation. As summarized in [Table 11](#), the results demonstrate the effectiveness of the heterogeneous strategy.

- On the NASA dataset, replacing the standard MLP with PIAF reduces the MAE from 0.0139 to 0.0066, representing a 52.5% reduction. This gain suggests that the Tanh branch in PIAF, with its saturation boundaries, captures the distinct voltage plateaus and abrupt degradation shifts typical of the NASA dataset, phenomena that the unbounded GeLU struggles to model accurately.
- On the Oxford dataset, PIAF achieves an RMSE of 0.0041 compared to 0.0068 for the baseline, corresponding to a 39.7% reduction.
- Even on the CALCE dataset, PIAF maintains its advantage, lowering the RMSE from 0.0264 to 0.0239.

This widespread improvement suggests that the “mixture of activations” strategy captures complex non-linearities that a single homogeneous activation function fails to model, thereby validating the feature richness of our design.

5.5.3. Micro-ablation II: Interaction analysis of attention modules

To elucidate the interaction dynamics between the diversity-enhancing FDFA and the efficiency-focused FLFA modules, specifically regarding whether their contributions are additive or synergistic, a

Table 11

Performance comparison of activation strategies (average results).

Dataset	Module variant	R ²	RMSE	MAE
NASA	Standard MLP (GeLU)	0.9612	0.0155	0.0139
	PIAF-MLP (Ours)	0.9880	0.0098	0.0066
Oxford	Standard MLP (GeLU)	0.9845	0.0068	0.0055
	PIAF-MLP (Ours)	0.9938	0.0041	0.0029
CALCE	Standard MLP (GeLU)	0.9826	0.0264	0.0197
	PIAF-MLP (Ours)	0.9856	0.0239	0.0169

stepwise ablation was conducted on the CALCE and NASA datasets. The AAT-1d was utilized as the baseline and progressively integrated the modules. The results are detailed in [Table 12](#).

Quantitative evidence of synergy. Experimental results, particularly on the CALCE dataset, suggest a super-additive synergistic effect:

- **Individual Gains:** On the CALCE dataset, adding the FDFA alone improves the MAE by 0.0019, reducing it from 0.0200 to 0.0181. Similarly, the independent addition of the FLFA yields a marginal improvement of 0.0005, lowering the MAE from 0.0200 to 0.0195. Consequently, the arithmetic sum of these isolated gains is 0.0024.
- **Combined Gain:** When both modules are integrated into the full AFF framework, the MAE drops significantly to 0.0169. This represents a total gain of 0.0031.
- **Conclusion:** Since the combined gain of 0.0031 exceeds the arithmetic sum of 0.0024, this indicates a positive synergistic interaction. The modules do not merely operate in parallel; rather, they mutually reinforce the predictive capability of the model.

Mechanistic explanation: Orthogonality and complementarity. This synergy arises because the two modules operate on nearly orthogonal feature dimensions, thereby generating complementary feature subspaces. The FDFA branch, utilizing non-linear ReLU pre-activation, targets the amplitude domain. It acts as a significance filter that isolates high-energy degradation signals from background noise. Conversely, the parallel FLFA branch, equipped with S-DConv and linear attention, focuses on the spatio-temporal structural domain to capture local-global dependencies regardless of signal magnitude. By fusing these distinct perspectives, specifically amplitude significance and temporal structure, the model achieves a comprehensive representation where the strengths of one branch compensate for the limitations of the other. This results in robustness that transcends simple summation.

5.6. Computational efficiency and feasibility analysis for BMS

To rigorously assess the suitability of SL-AgentNet for resource-constrained embedded BMS, we conducted a comprehensive benchmark focusing on four critical engineering metrics:

- **Floating Point Operations (FLOPs):** Represents the theoretical computational complexity. In this study, we report the *cumulative*

Table 12

Interaction ablation of FDFA and FLFA modules (Avg. results).

Dataset	Attention variant	R^2	RMSE	MAE	MAPE
CALCE	Baseline (AAT-1d)	0.9828	0.0256	0.0200	0.0349
	+ FDFA (Diversity)	0.9850	0.0242	0.0181	0.0333
	+ FLFA (Global-Local)	0.9841	0.0249	0.0195	0.0326
	Full AFF (Combined)	0.9856	0.0239	0.0169	0.0288
NASA	Baseline (AAT-1d)	0.8933	0.0259	0.0237	0.0325
	+ FDFA (Diversity)	0.9791	0.0117	0.0086	0.0113
	+ FLFA (Global-Local)	0.9837	0.0103	0.0074	0.0094
	Full AFF (Combined)	0.9880	0.0098	0.0066	0.0085

FLOPs required to process a complete charging cycle (macro-scale cost), serving as a proxy for total energy consumption.

- **Number of Parameters (Params):** Indicates the model's architectural complexity. Fewer parameters reduce memory access frequency and the risk of overfitting on small datasets.
- **Model Size (Storage):** The actual Flash memory footprint required to store model weights, which is a stringent constraint for automotive microcontrollers (typically < 1 MB).
- **Inference Latency:** The real-world execution time required to process a single input sequence (micro-scale response). This determines whether the model can operate within the real-time scheduling slot (e.g., 10 ms task cycle) of the BMS.

5.6.1. Experimental setup for latency measurement

Unlike theoretical FLOPs, inference latency is typically considered a primary determinant of real-time feasibility on embedded controllers. A standardized benchmark environment was established using a CPU to simulate the sequential processing characteristic of BMS MCUs, adhering to the following protocols:

- **Device & Process:** Measurements were executed on a CPU with a batch size of 1 to mimic the stream-processing nature of BMS operations. The protocol comprised 10 warm-up runs followed by 1000 active measurement cycles to promote statistical robustness; only forward inference time was recorded, excluding data I/O overhead.
- **Input Constraints:** The input dimensions were fixed at $(1, N, C)$. While Transformer and LSTM baselines utilized a context window of $N = 5$, SL-AgentNet employed a restricted window of $N = 4$ due to the architectural configuration of the Agent mechanism. Achieving improved accuracy with this reduced receptive field— $N = 4$ versus $N = 5$ —highlights the enhanced efficiency of SL-AgentNet's feature extraction capabilities.

5.6.2. Inference latency and real-time capability

Table 13 indicates that SL-AgentNet achieves a sub-millisecond inference latency of 0.7656 ms alongside a throughput of 1306 samples/s. Furthermore, the model demonstrates the lowest latency standard deviation of 0.0907 ms among all tested architectures. This minimal variance suggests enhanced temporal stability—an important factor for supporting deterministic behavior in safety-critical Real-Time Operating Systems (RTOS).

Although the LSTM baseline exhibits marginally lower latency (≈ 0.69 ms) due to minimal recurrent overhead in short sequences, SL-AgentNet performs comparably to the efficient CNN-LSTM (≈ 0.74 ms) and outperforms Transformer-based models (≈ 1.0 – 1.1 ms). These results indicate that the linear-complexity FLFA module mitigates the computational burden typically inherent in attention mechanisms.

5.6.3. Distinguishing cumulative FLOPs from low latency

A potential ambiguity arises when contrasting the high FLOPs count (589 M) with the minimal latency (0.77 ms). It is important to clarify the physical definition of these metrics within our experimental framework:

- **FLOPs (Macro-scale Cost):** The FLOPs values reported in Table 13 quantify the cumulative computational cost for processing a complete battery charge/discharge cycle. Derived from a representative Oxford dataset cycle comprising thousands of time steps, this metric evaluates the total energy consumption for a full diagnostic event.
- **Latency (Micro-scale Response):** Latency measures the inference time for a single sliding window input (e.g., $N = 4$), representing the instantaneous response time required by the BMS controller per execution step.

Consequently, high cumulative FLOPs do not contradict low instantaneous latency. Given that SOH estimation is a low-frequency prognostic task—typically executed once per cycle or at minute-level intervals—the cumulative load is distributed over extended durations. This renders the 0.77 ms per-step latency likely compatible with automotive MCU background task scheduling.

5.6.4. The efficiency-accuracy trade-off

SL-AgentNet distinguishes itself through a strategic balance between resource consumption and estimation reliability. While the CNN-LSTM baseline offers the lowest computational cost (115 M FLOPs) and marginally faster inference (0.74 ms), this lightweight nature compromises representation capacity, resulting in a relatively high RMSE of 0.0100. In safety-critical BMS applications, an estimation error of 1% is frequently deemed the threshold of unreliability, potentially necessitating conservative battery utilization or obscuring rapid degradation signals.

Conversely, SL-AgentNet exchanges a manageable increase in computational load—remaining well within the sub-millisecond safety margin of 0.77 ms—for enhanced performance. By reducing the RMSE to 0.0042, our model achieves a 58% error reduction relative to CNN-LSTM. This trade-off is justifiable for modern electric vehicles: the incremental increase in storage (40 KB vs. 15 KB) and FLOPs is negligible for automotive-grade MCUs, whereas the improved accuracy contributes to extended battery lifespan and safer operational limits. Thus, SL-AgentNet provides an effective solution where precision is paramount, yet hardware constraints remain respected.

5.6.5. Implementation challenges and hardware validation

Deploying deep learning models on embedded systems requires addressing specific hardware limitations, particularly regarding non-linear operations.

Floating-point unit (FPU) limitations. The proposed PIAF module utilizes heterogeneous activation functions (e.g., Tanh, GeLU) involving exponential calculations (e^x). On low-power automotive MCUs, these operations can be computationally expensive and may induce latency jitter.

Hardware-in-the-loop validation. To validate the feasibility of the proposed framework, we conducted deployment experiments on a **TI Cortex-M4** microcontroller at **Sichuan SEVCLAB Testing Technology Co., Ltd.** (a subsidiary of the Tsinghua University Prof. Ouyang Minggao Academician Workstation).

- **Look-Up Table (LUT) Optimization:** We implemented the exponential-heavy activations using LUTs with piecewise linear interpolation. This is a standard industry practice to balance precision and speed.
- **Experimental Results:** Comparative testing between the PC-based floating-point inference and the MCU-based LUT implementation confirms that the optimization significantly reduces cycle counts. Crucially, the SOH estimation discrepancy (accuracy loss) introduced by this approximation is controlled strictly within **0.3%**, which is negligible for BMS applications.

Table 13

Comparison of computational efficiency, model complexity, and inference latency.

Model	Model complexity & Cost			Runtime performance (CPU, Batch = 1)		
	Params	Size (KB)	FLOPs (M)*	Latency (ms)	Std (ms)	Throughput (samples/s)
SL-AgentNet	6,114	40.92	589.45	0.7656	0.0907	1306.09
Transformer	17,569	90.53	126.14	1.0086	0.1370	991.45
LSTM	99,585	392.15	1807.94	<u>0.6929</u>	0.0967	<u>1443.28</u>
CNN-Transformer	43,169	212.06	1805.76	1.1395	0.1391	877.58
CNN-LSTM	<u>3,041</u>	<u>15.58</u>	<u>115.20</u>	0.7423	0.1001	1347.18

Note: **Bold** indicates the proposed SL-AgentNet; Underline indicates the best result in each column.Latency is measured on a standard CPU. Input $N = 4$ for SL-AgentNet, $N = 5$ for others.

*FLOPs represent the cumulative computational cost over a complete battery charging cycle (approx. thousands of steps).

Table 14Dataset summary metrics (Mean and 95% Confidence Interval). 90% prediction intervals were generated using conformal prediction (q_{hat}).

Dataset	MAE (95% CI)	RMSE (95% CI)	Mean PICP (95% CI)	MPIW	q_{hat}
NASA	0.0049 [0.0035, 0.0062]	0.0085 [0.0062, 0.0108]	0.9560 [0.9344, 0.9776]	0.0295	0.0148
CALCE	0.0190 [0.0172, 0.0210]	0.0261 [0.0220, 0.0311]	0.8995 [0.8492, 0.9415]	0.0779	0.0390

Table 15Statistical calibration verification using Exact Binomial Test on aggregate samples ($H_0 : \text{PICP} = 0.90$).

Dataset	N_{total}	Covered (k)	Observed PICP	p -value	Conclusion
NASA	408	391	95.83%	2.16×10^{-5}	Conservative (Safe)
CALCE	2702	2432	90.01%	1.00	Precisely Calibrated

Table 16

Paired t-test results comparing SL-AgentNet against baselines. Positive Mean Diff indicates SL-AgentNet outperforms baseline.

Significance threshold $\alpha' \approx 0.003$ (Bonferroni corrected).

Dataset	Baseline	Metric	Mean Diff.	t -statistic	p -value	Cohen's d [95% CI]
NASA	LSTM	MAE	+0.0107	20.97	< 0.001***	1.04 [0.90, 1.21]
	Transformer	MAE	+0.0089	16.48	< 0.001***	0.82 [0.75, 0.89]
	LSTM	RMSE ^a	+0.0090	15.27	< 0.001***	7.64 [2.85, 12.4]
	Transformer	RMSE ^a	+0.0089	4.15	0.026 (ns)	2.07 [0.15, 4.00]
CALCE	Transformer	MAE	+0.0136	30.66	< 0.001***	0.59 [0.56, 0.62]
	LSTM	MAE	-0.0014	-5.86	< 0.001***	-0.11 [-0.15, -0.07]
	Transformer	RMSE ^a	+0.0194	4.66	0.019 (ns)	2.33 [0.28, 4.38]
	LSTM	RMSE ^a	-0.0025	-1.15	0.332 (ns)	-0.58 [-1.84, 0.68]

(ns) Not significant at adjusted threshold.

^a RMSE analysis based on aggregated cell-level metrics ($N = 4$).*** Significant at $p < 0.003$ (Bonferroni adjusted).

Stream processing. Additionally, unlike offline batch training, the BMS operates on a continuous stream. Our latency benchmarks (Section 5.6.2) were specifically conducted with a batch size of 1. The verified throughput (1306 samples/s) far exceeds the typical 1–10 Hz sampling rate of voltage sensors, ensuring no data backlog risks in real-time operation.

6. Statistical significance and uncertainty quantification

Statistical significance testing and conformal prediction analysis were incorporated to assess the reliability of SL-AgentNet beyond standard error metrics.

6.1. Statistical significance and effect size

Paired t-tests were performed to verify performance differences, with the Bonferroni correction applied to maintain robustness against multiple comparisons ($\alpha' \approx 0.003$). The results are detailed in Table 16.

NASA dataset (Strong generalization). SL-AgentNet shows strong performance on the NASA dataset. Compared to both LSTM and Transformer baselines, our model achieves significant reductions in MAE ($p < 0.001$) with large effect sizes (Cohen's $d > 0.8$). Regarding RMSE, while SL-AgentNet achieves a lower mean error with a large effect size ($d = 2.07$

vs. Transformer), the p -value (0.026) did not reach the Bonferroni threshold. This suggests that while the improvement magnitude is practically significant, statistical power is limited by the small sample size ($N = 4$ cells).

CALCE dataset (Engineering equivalence). On the CALCE dataset, SL-AgentNet maintains a significant lead over the Transformer in MAE ($p < 0.001$). Compared to the LSTM baseline, the analysis reveals:

1. **RMSE Equivalence:** No statistically significant difference was observed ($p = 0.332$), indicating comparable performance in minimizing large squared errors.
2. **MAE Engineering Equivalence:** Although LSTM shows a slight statistical preference in MAE ($p < 0.001$), the mean difference is merely -0.0014 (0.14% SOH), which is negligible compared to typical sensor noise.

Thus, SL-AgentNet demonstrates engineering equivalence to LSTM while offering the advantage of valid uncertainty quantification.

6.2. Prediction uncertainty and calibration

90% Prediction Intervals (PIs) were constructed using conformal prediction. The comprehensive performance metrics, including error

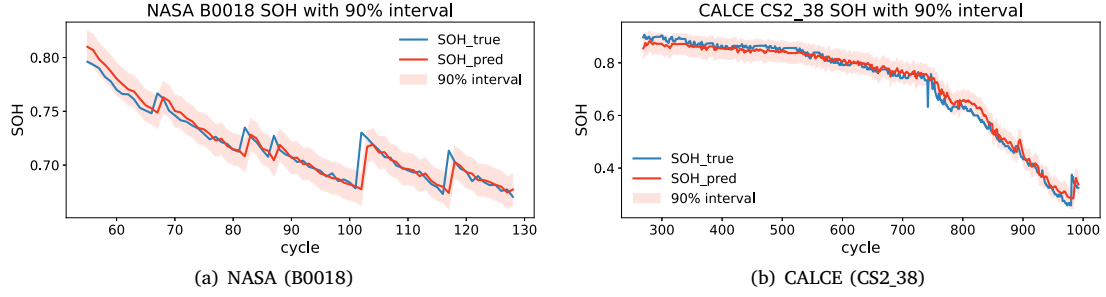


Fig. 13. Prediction Uncertainty Visualization. The red line represents ground truth, and the shaded area is the **90% Prediction Interval (PI)**. B0018 shows strong capacity regeneration, while CS2_38 exhibits rapid non-linear decay. SL-AgentNet effectively brackets these challenging fluctuations.

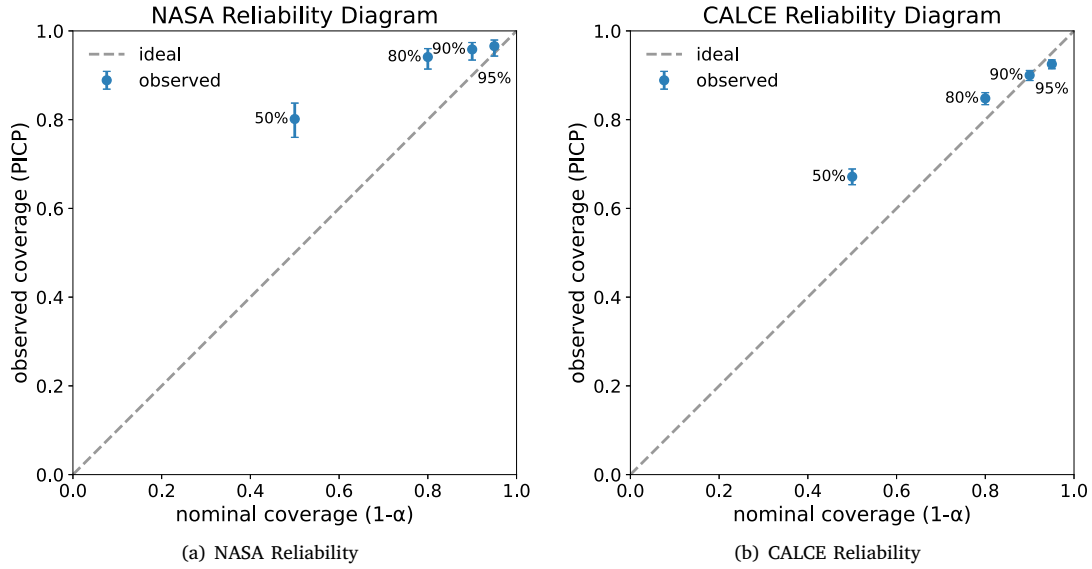


Fig. 14. Reliability Diagrams. Calibration curves across confidence levels (50%, 80%, 90%, 95%). Error bars indicate 95% Wilson score intervals. (a) NASA shows consistent conservative calibration (points above diagonal). (b) CALCE shows precise calibration at the 90% operating point (error bar intersects the diagonal).

margins and uncertainty quantification results, are summarized in Table 14. The calibration quality was further verified via statistical testing (Table 15) and visual analysis (Fig. 14). The assessment relies on the probabilistic metrics (PICP, MPIW, and ACE) defined in Section 5.1.

Calibration analysis. Using the **Exact Binomial Test** ($H_0 : \text{PICP} = 0.90$):

- **NASA (Conservative Safety):** The aggregate coverage is 95.83% ($p < 0.001$), indicating a safety-conservative bias that supports higher fault tolerance for degradation risks.
- **CALCE (Precise Calibration):** The aggregate coverage is 90.01% ($p = 1.00$), confirming no significant deviation from the nominal level and demonstrating precise calibration.

Reliability diagrams (Fig. 14) further support these findings, with CALCE closely following the ideal diagonal ($\text{ACE}=0.061$) and NASA showing consistent conservative behavior ($\text{ACE}=0.129$). This framework offers a stable safety margin (constant width intervals, Fig. 13) beneficial for robust BMS deployment.

7. Conclusion

This study presents a practical, computationally efficient framework for battery SOH estimation, designed to address the stringent operational constraints of vehicular BMS. By recognizing the limitations of

direct measurement and the high computational cost of standard deep learning models, a dual-pronged solution comprising systematic feature selection and a lightweight network architecture was developed.

The proposed SL-AgentNet is characterized by its linear computational complexity. By integrating the AFF module with the Fast Local-Global Fusion Attention (FLFA) mechanism, the model captures long-range degradation dependencies while addressing the quadratic computational overhead ($O(N^2)$) associated with traditional Transformers. Furthermore, the incorporation of FDFA and PIAF modules enhances feature diversity and non-linear representation, facilitating high estimation accuracy within a lightweight architecture.

Experimental results on the Oxford, NASA, and CALCE datasets indicate that SL-AgentNet demonstrates enhanced performance compared to baseline models, including the LSTM and CNN-Transformer. Specifically, the model achieves significant accuracy gains, with MAE reductions reaching 74.78% on the Oxford dataset compared to the CNN-Transformer baseline. Crucially, this competitive performance is realized with a parameter count of only 6114 and a storage footprint of 40.92 KB. Hardware-in-the-loop validation on a TI Cortex-M4 MCU has further confirmed the model's engineering feasibility, where LUT optimization ensured real-time responsiveness with a negligible accuracy loss of within 0.3%.

Despite these advancements, limitations remain regarding the reliance on stable constant-current segments for feature extraction. Future work aims to extend the framework to encompass a wider range

of battery chemistries (e.g., LFP) and extreme dynamic operating conditions to further broaden its applicability in real-world scenarios.

CRedit authorship contribution statement

Huajiang Wang: Writing – review & editing, Writing – original draft, Conceptualization. **Hao Li:** Formal analysis, Data curation, Conceptualization. **Zhengxing He:** Methodology, Investigation, Funding acquisition. **Xuan Chen:** Methodology, Investigation. **Hanglin Liu:** Software, Resources, Project administration. **Chao Chen:** Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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